
Towards Understanding Modality Interaction in Multimodal Language Models via Partial Information Decomposition

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Abstract

Understanding how multimodal large language models use different modalities is important for reliable reasoning. We employ Partial Information Decomposition (PID) as a decision-level lens and introduce *Sensory PID*, a conditional formulation that conditions on language and separates unique, redundant, and synergistic contributions from video and audio. Applied to omni-modal models, Sensory PID reveals a sensory synergy bottleneck: even on audio–visual fusion tasks, decisions remain dominated by modality-unique information, with stronger reliance on vision. Modality-shuffling interventions support this asymmetry, while layer-wise analysis reveals a visual-first computation pattern and instruction perturbations show that late-stage sensory fusion is conditioned by language. Beyond diagnosis, PID-guided sample reweighting provides initial evidence that local diagnostic signals can improve multimodal reasoning and grounding performance. As reference validation, our vision–language analysis broadly corroborates previously reported decision-level PID patterns across tasks, models, interventions, and layers.

1. Introduction

Multimodal large language models (MLLMs) are moving from perception systems toward *decision-making agents* in scientific analysis, medical support, and embodied interaction (Yin et al., 2024; Jin et al., 2025; Wang et al., 2025).

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By integrating vision, audio, and language within unified generative backbones, these models can ground abstract instructions in rich sensory inputs and produce context-sensitive responses (Liang et al., 2024; Deng et al., 2025; Zhou et al., 2022). Yet current evaluation paradigms still primarily assess *what* models predict, while providing limited insight into *how* different modalities contribute to those predictions (Pan et al., 2024; Deitke et al., 2025).

A growing body of work has sought to go beyond outcome accuracy through representation analysis, cross-modal alignment, attention-based inspection, and modality ablation (Yan et al., 2025; Lou et al., 2025; Yoon et al., 2025; Jiang et al., 2025; Li et al., 2025). These approaches provide complementary views of how modalities are encoded and how predictions respond to perturbations. Partial Information Decomposition (PID) offers another perspective by separating multimodal information into unique, redundant, and synergistic components. Our prior work used PID to characterize multimodal information structure and inform representation alignment (Fang et al., 2026). More recently, Xiu et al. (2026) take a first step to extend PID to decision-level analysis of large vision–language models, revealing systematic information-use patterns across tasks, model families, interventions, and network depth.

These developments motivate a broader question: how do MLLMs use different modalities when those modalities play distinct roles in decision-making? This question becomes especially important in omni-modal systems, where language specifies what the model should resolve while video and audio provide sensory evidence. We therefore introduce *Sensory PID* (Figure 1), a conditional decision-level decomposition of the sensory information gain $I(Y; V, A \mid T)$. By conditioning on language, Sensory PID separates video-unique, audio-unique, redundant, and synergistic information while distinguishing task specification from sensory evidence.

Applied to omni-modal models, Sensory PID reveals a *sensory synergy bottleneck*: even on questions designed to require joint audio–visual reasoning, decision-relevant information remains dominated by modality-unique contributions. Independent video and audio shuffling supports a visually dominant decision profile. Layer-wise analysis

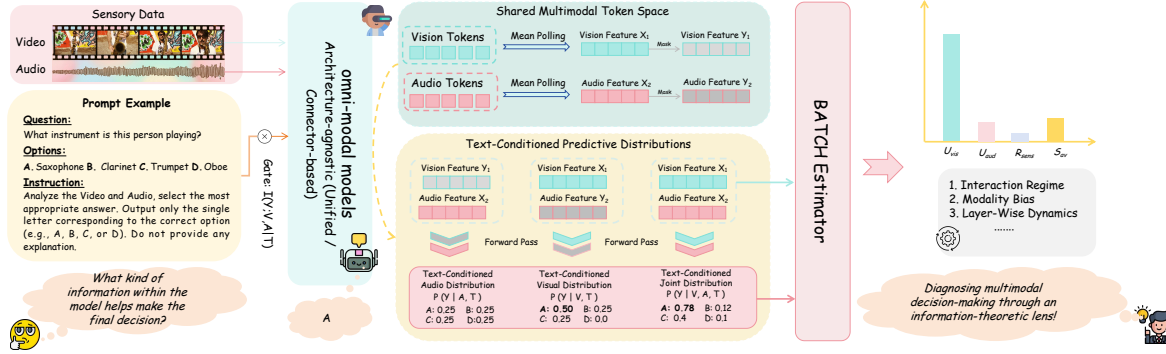


Figure 1. **The Sensory PID Framework.** (Left) Omni-modal models are analyzed by treating the text prompt as a conditioning signal for video–audio interaction. (Right) Applying the BATCH estimator in the shared token space decomposes decision-level information into Unique (U_{vis} , U_{aud}), Redundant (R_{sens}), and Synergistic (S_{av}) components.

further reveals a *visual-first* pattern in which visual-unique information becomes decision-relevant before auditory information and audiovisual synergy, while controlled instruction perturbations show that late-stage sensory fusion varies with linguistic task demand.

Beyond diagnosis, we ask whether PID-derived signals can guide model adaptation. In a separate vision–language proof of concept, motivated by our prior use of PID to inform multimodal intervention design (Fang et al., 2026), we use BATCH-induced local diagnostics to reweight samples during lightweight LoRA fine-tuning. The policy upweights samples with high fusion potential but limited current synergy and downweights language-shortcut samples. The results provide initial evidence that local information-decomposition signals can improve reasoning and grounding over uniform fine-tuning and matched reweighting baselines, while the contrast across model profiles suggests that the benefit is profile-dependent.

For reference validation, we also apply the analogous two-source decision-level PID analysis to 20 vision–language models across six benchmarks. These experiments broadly corroborate previously reported task-, model-, intervention-, and layer-dependent patterns (Xiu et al., 2026), while extending the reference evaluation to MMStar, MMMU, and Qwen3-VL.

We summarize our main contributions as follows:

Conditional Sensory PID for omni-modal models. We introduce a conditional decomposition of $I(Y; V, A | T)$ that treats language as task context and separates the decision-relevant information contributed individually and jointly by video and audio.

Omni-modal sensory interaction findings. Across the evaluated omni-modal models, Sensory PID reveals a sensory synergy bottleneck, intervention-supported visual dom-

inance, visual-first information dynamics, and instruction-conditioned sensory fusion.

PID-guided sample reweighting. In a separate VLM proof of concept, we formulate a PID-guided sample-reweighting policy that converts BATCH-induced local information-decomposition diagnostics into sample rankings and training weights. The resulting experiments provide initial evidence of reasoning and grounding gains in joint-modality-reliant models while exposing a profile-dependent boundary case.

2. Methodology

Our methodology has two complementary components. First, we introduce *Sensory PID*, a conditional decision-level decomposition for characterizing how video and audio contribute to an omni-modal model’s decision under a textual task specification. Second, we formulate a PID-guided sample-reweighting policy that converts local information-decomposition diagnostics into sample-ranking and training signals for model adaptation. The former provides an analytical framework for language-conditioned sensory interaction, while the latter translates information decomposition from diagnosis into a targeted training intervention.

2.1. Conditional Sensory PID for Omni-Modal Models

Omni-modal models process a video stream V , an audio stream A , and a textual instruction T to produce a response Y . Our goal is to characterize how the two sensory streams contribute individually and jointly to the model’s decision *under a fixed linguistic task specification*. A full three-source PID is possible in principle, but introduces a combinatorial number of partial-information terms and treats language symmetrically with the sensory inputs. In instruction-following settings, however, these modalities play different roles: we treat T as the conditioning context specifying the

task, while V and A provide the sensory evidence relevant to that task.

We therefore introduce *Sensory PID*, which decomposes the conditional sensory information gain

$$I(Y; V, A | T) = U_{\text{vis}} + U_{\text{aud}} + R_{\text{sens}} + S_{\text{av}}, \quad (1)$$

where U_{vis} and U_{aud} denote information uniquely attributable to video and audio, respectively; R_{sens} denotes sensory information redundantly available from either stream; and S_{av} denotes information available only from their joint observation, conditioned on the instruction.

The conditioning structure is central to the formulation. Rather than treating text as a third PID source symmetric with video and audio, Sensory PID keeps the instruction fixed and asks how the available sensory evidence contributes to the resulting decision. This separates task specification from sensory contribution and changes the object of decomposition from generic multi-source interaction to language-conditioned video–audio interaction.

Operationally, the analysis is defined through three model-induced, text-conditioned predictive distributions:

$$\begin{aligned} p_{\theta}(Y | V, A, T), \quad p_{\theta}(Y | V, T), \\ p_{\theta}(Y | A, T). \end{aligned} \quad (2)$$

corresponding respectively to the full sensory prediction and the two partial-sensory conditions. The same instruction T is retained in all three cases, so their differences reflect the sensory evidence available to the model. Section 2.2 describes how these distributions and the resulting Sensory PID components are estimated in practice.

2.2. Estimating Sensory PID in Omni-Modal Models

Estimating Sensory PID in modern omni-modal models requires addressing three practical issues: sensory inputs are represented by variable-length token sequences, jointly trained models do not always provide native partial-sensory inference paths under a fixed instruction, and restricted candidate distributions can become unstable when one sensory stream is removed. We therefore construct a training-free estimation pipeline around the BATCH estimator (Liang et al., 2023), as illustrated in Figure 1.

Sensory source representations. Omni-modal architectures map video, audio, and language into a shared multi-modal representation space. Because BATCH operates on fixed-dimensional continuous source variables, we summarize the video and audio token sequences using mean pooling after their projection into this shared space. This yields sensory source representations $X_V, X_A \in \mathbb{R}^d$, while the textual instruction remains present as conditioning context

throughout inference. Mean pooling provides a parameter-free, sample-aligned summary of each sensory stream; sensitivity to alternative pooling strategies is examined in Appendix G.

Text-conditioned sensory predictions. For each sample (V, A, T) , the full omni-modal forward pass produces $p_{\theta}(Y | V, A, T)$. To estimate the two partial-sensory conditions required by Eq. 1, we additionally construct $p_{\theta}(Y | V, T)$ and $p_{\theta}(Y | A, T)$ while keeping the same instruction T . Across the profiling set $\mathcal{D}_{\text{prof}}$, these predictions define the model-induced decision distributions used for Sensory PID estimation.

Calibrated sensory masking. Because current omni-modal models do not always expose native video–text or audio–text execution paths, we approximate these partial-sensory conditions through representation corruption. To estimate $p_{\theta}(Y | V, T)$, we replace the audio embeddings while retaining video and text; conversely, to estimate $p_{\theta}(Y | A, T)$, we replace the video embeddings while retaining audio and text. For a masked sensory modality $m \in \{V, A\}$,

$$\tilde{E}_m[\ell] \sim \mathcal{N}(\mu_m, \text{diag}(\sigma_m^2)), \quad \ell = 1, \dots, L_m, \quad (3)$$

where μ_m and σ_m are per-dimension statistics estimated over $\mathcal{D}_{\text{prof}}$. Matching the empirical feature statistics controls the perturbation scale while removing instance-specific information from the masked sensory stream.

Sensory PID estimation. We apply BATCH (Liang et al., 2023) to the video and audio source representations together with the text-conditioned decision distributions. BATCH learns a Sinkhorn-normalized coupling satisfying the required source–target marginal constraints, from which U_{vis} , U_{aud} , R_{sens} , and S_{av} are recovered. The instruction therefore specifies the decision context throughout the estimation, while the decomposition characterizes how video and audio contribute information within that context.

Output stabilization. For multiple-choice evaluation, each predictive distribution is restricted to the candidate set $\mathcal{C}$ using selective softmax. Prior decision-level LVLM PID analysis uses a hard confidence gate: when the probability mass assigned to valid candidates falls below a threshold τ , the corresponding candidate distribution is replaced by a uniform distribution before soft aggregation (Xiu et al., 2026).

This stabilization rule is not directly aligned with our conditional sensory setting. Sensory PID jointly relies on the full-sensory prediction $p_{\theta}(Y | V, A, T)$ and the two partial-sensory predictions $p_{\theta}(Y | V, T)$ and $p_{\theta}(Y | A, T)$. Because differences among these predictive conditions directly

enter the estimation of sensory-unique and audiovisual-synergistic information, independently replacing a distribution once its confidence crosses a hard threshold can introduce an abrupt branch-specific change that is unrelated to the underlying sensory evidence.

We therefore use a continuous confidence-dependent stabilization. For each sample i and predictive condition c , let

$$m_i^{(c)} = \sum_{y \in \mathcal{C}} S_{\text{orig},i}^{(c)}(y) \quad (4)$$

denote the pre-renormalization probability mass assigned to valid candidates. We define

$$\rho_i^{(c)} = \min \left\{ 1, \frac{m_i^{(c)}}{\tau} \right\}, \quad (5)$$

and stabilize the corresponding predictive distribution as

$$\tilde{p}_i^{(c)}(y) = \rho_i^{(c)} p_i^{(c)}(y) + \left(1 - \rho_i^{(c)}\right) \frac{1}{|\mathcal{C}|}. \quad (6)$$

Predictions above the confidence threshold are therefore unchanged, whereas lower-confidence predictions are progressively shrunk toward the uniform distribution rather than being replaced discontinuously.

For the target marginal, we retain the soft-aggregation principle used in prior decision-level LVLM PID analysis (Xiu et al., 2026):

$$\bar{p}(y) = \frac{1}{|\mathcal{D}_{\text{prof}}|} \sum_{j \in \mathcal{D}_{\text{prof}}} \tilde{p}_j^{(VAT)}(y), \quad (7)$$

and keep this marginal fixed during BATCH optimization (Liang et al., 2023). Thus, our implementation shares the soft-aggregation principle with prior LVLM PID analysis but differs in how low-confidence predictive conditions are stabilized before aggregation. (Liang et al., 2023).

Estimator provenance and vision–language reference. BATCH, mean pooling, embedding-space corruption, confidence regularization, and soft aggregation are implementation choices for estimating PID quantities rather than methodological contributions of this work. Closely related combinations of these choices have also been reported in prior decision-level LVLM analysis (Xiu et al., 2026). Our contribution in the omni-modal setting lies in formulating and applying conditional Sensory PID for language-conditioned video–audio interaction, rather than in these individual estimator components.

For the vision–language reference analysis and the reweighting method introduced in Section 2.3, we use the analogous two-source vision–language formulation; its definition and implementation details are provided in Appendix A.1. The underlying two-source VLM PID formulation and estimator are not treated as methodological contributions of this work.

2.3. PID-Guided Reweighting from Local Diagnostics

From diagnosis to intervention. Beyond using information decomposition as an analytical tool, we ask whether its local diagnostic signals can guide model adaptation. Motivated by our prior use of PID to inform multimodal intervention design (Fang et al., 2026), we formulate a PID-guided sample-reweighting policy for vision–language models. The policy is designed to distinguish samples with unrealized cross-modal fusion potential from those dominated by language-side shortcuts. Importantly, the contribution here is not a new PID estimator, but the conversion of BATCH-induced local diagnostic signals into a structured sample-ranking and reweighting policy.

Local information-decomposition diagnostics. Standard sample-selection criteria conflate different sources of difficulty. Accuracy-based mining identifies hard examples but cannot distinguish weak multimodal fusion from missing knowledge, while modality-ablation scores measure dependence on a modality without separating unique from synergistic information.

For each training sample i , BATCH induces local additive contributions s_i , $u_{\text{txt},i}$, $u_{\text{vis},i}$, and r_i whose averages recover the aggregate PID terms. These quantities are used as BATCH-induced diagnostic contributions rather than axiomatic pointwise PID atoms; their full derivation and stabilization procedure are provided in Appendix A.4.

Because the local contributions need not be non-negative, we apply non-negative clipping, $[z]_+ = \max(z, 0)$, and define the per-sample information mass

$$I_i^+ = [s_i]_+ + [u_{\text{vis},i}]_+ + [u_{\text{txt},i}]_+ + [r_i]_+. \quad (8)$$

We then define the *synergy ratio* and *shortcut score*

$$\text{SR}_i = \frac{[s_i]_+}{I_i^+ + \epsilon}, \quad \text{SC}_i = \frac{[u_{\text{txt},i}]_+}{I_i^+ + \epsilon}, \quad (9)$$

where $\epsilon = 10^{-8}$. To reduce dependence on individual mini-batch composition, the local contributions are averaged over $K = 50$ random batch contexts before constructing these scores.

Fusion-potential and synergy-gap scoring. The frozen VLM additionally provides three predictive distributions: $p_v^{(i)} = P(Y | V_i)$, $p_t^{(i)} = P(Y | T_i)$, and $p_{vt}^{(i)} = P(Y | V_i, T_i)$. We define the *fusion potential*

$$\text{FP}_i = \left[\min \left\{ H(p_v^{(i)}), H(p_t^{(i)}) \right\} - H(p_{vt}^{(i)}) \right]_+, \quad (10)$$

which measures how much the joint prediction sharpens relative to the better of the two partial-input predictions.

Fusion potential alone does not indicate whether the model already uses the two modalities synergistically. We therefore combine it with the local PID profile and define

$$\text{GapScore}_i = (1 - \text{SR}_i)(1 - \text{SC}_i) \text{FP}_i, \quad (11)$$

which prioritizes samples with high apparent fusion potential, limited current synergy, and limited text-unique shortcut reliance.

PID-guided sample reweighting. The resulting policy has two complementary priorities. Samples with the largest SC_i are treated as *language-shortcut* examples and down-weighted, while samples with the largest GapScore_i among the remaining examples are treated as *synergy-gap* examples and upweighted:

$$\begin{aligned} \mathcal{S}_{\text{short}} &= \text{TopK}_{K_s}(\text{SC}_i), \\ \mathcal{S}_{\text{gap}} &= \text{TopK}_{K_g}(\text{GapScore}_i \mid i \notin \mathcal{S}_{\text{short}}). \end{aligned} \quad (12)$$

We assign $w_i = 3.0$ to synergy-gap samples, $w_i = 0.5$ to language-shortcut samples, and $w_i = 1.0$ otherwise. The resulting weights are used during lightweight LoRA fine-tuning. Algorithmic details and the effective training distribution are reported in Appendix H.

3. Experimental Setup

Our evaluation is organized around two complementary roles. The *primary* experiments study language-conditioned sensory interaction in omni-modal models using Sensory PID, including modality-specific interventions, layer-wise analysis, and controlled instruction perturbations. Separately, we retain a vision-language evaluation as a reference check for the decision-level PID pipeline. As this reference analysis primarily corroborates previously reported LVLM patterns (Xiu et al., 2026), we keep its presentation concise in the main text and defer the detailed results to appendix.

Omni-modal models. We evaluate three model configurations from two omni-modal families: Qwen2.5-Omni 3B and 7B (Xu et al., 2025), and VITA-1.5 7B (Fu et al., 2025). These models support joint video, audio, and text inputs and provide coverage across distinct model families and parameter scales.

Omni-modal benchmark and sensory subsets. Our primary omni-modal evaluation uses MUSIC-AVQA (Li et al., 2022), an audio-visual question-answering benchmark based on music videos. To distinguish sensory requirements, we construct three balanced subsets from the dataset metadata: *Audio-Focused*, containing questions whose answers depend primarily on auditory evidence; *Visual-Focused*, emphasizing visual objects or actions; and *AV-Fusion*, containing questions designed to require joint audio-visual

reasoning. The AV-Fusion subset serves as our primary testbed for sensory synergy, while the Audio-Focused and Visual-Focused subsets provide reference conditions for modality-specific information use. Detailed dataset statistics and construction criteria are provided in Appendix E.

Sensory evaluation and intervention protocol. All models are evaluated zero-shot using standardized prompts and deterministic greedy decoding. For Sensory PID estimation, the same textual instruction is retained across the full and partial-sensory conditions described in §2.1, so that the decomposition compares sensory evidence under a fixed linguistic task context.

We complement the information decomposition with three analyses that probe different aspects of sensory interaction. First, we independently shuffle video or audio streams across samples, breaking the correspondence of one sensory modality while preserving its sample-level marginal distribution. This provides a behavioral measure of the model’s sensitivity to correctly aligned visual and auditory evidence. Second, we trace Sensory PID components across transformer depth to examine when video-, audio-, and jointly derived information become decision-relevant. Third, we hold the video and audio inputs fixed while modifying the textual instruction to reduce its demand for joint audio-visual reasoning, allowing us to examine whether sensory fusion changes with linguistic task specification.

Vision-language reference evaluation. As a separate reference validation, we evaluate 20 vision-language models from 7 model families across six benchmarks: MMBench, MMStar, MMMU, PMC-VQA, POPE, and Reefknot. We additionally obtain a text-only condition by removing the image input and measure Δ_{vision} as the corresponding accuracy drop. These experiments serve as a reference check and broadly recover the task-, model-, intervention-, and layer-dependent PID patterns reported in prior LVLM analysis (Xiu et al., 2026), rather than constituting the primary empirical contribution of this work. Benchmark details and full per-model results are provided in Appendices E and F. The separate training setup for the PID-guided reweighting experiments is described in §5.

4. Sensory Interaction in Omni-Modal Models

Vision-language reference validation. As a reference check, we also apply the analogous decision-level PID analysis to 20 vision-language models across six benchmarks. The resulting task-, model-, intervention-, and layer-dependent patterns are broadly consistent with prior LVLM PID analysis (Xiu et al., 2026): benchmarks separate into relatively synergy-heavy and text-unique-heavy regimes, family-level modality-use tendencies are reflected in vision-

removal sensitivity, and the previously reported late-stage shift toward cross-modal synergy also appears in our layer-wise analyses on MMStar and MMMU. Because these analyses primarily serve as reference validation, we defer the complete results, correlations, and layer-wise trajectories to Appendix F and focus below on sensory interaction in omni-modal models.

4.1. Sensory Synergy Bottleneck and Visual Dominance

We first examine how video and audio contribute to decisions on the AV-Fusion subset of MUSIC-AVQA, where questions are designed to emphasize joint audio-visual reasoning. If the two sensory streams are strongly integrated at the decision level, these questions should exhibit substantial sensory synergy. Instead, Sensory PID reveals a pronounced imbalance between modality-unique and jointly available sensory information.

4.1.1. THE SENSORY SYNERGY BOTTLENECK

Table 1 shows that sensory synergy S_{av} remains a relatively small component across the evaluated omni-modal models, even on the AV-Fusion subset. Rather than deriving most decision-relevant sensory information from the joint observation of video and audio, the models remain dominated by modality-unique information. We refer to this pattern as a *sensory synergy bottleneck*: supporting multiple sensory streams architecturally does not necessarily yield strong decision-level audiovisual integration.

The decomposition also reveals a consistent asymmetry between the two sensory modalities. Unique visual information U_{vis} generally exceeds unique auditory information U_{aud} , indicating stronger reliance on visual evidence. This visual skew motivates the intervention analysis below, where we test whether the information imbalance is accompanied by asymmetric sensitivity to disrupting the two sensory streams.

Finding 1. Across the evaluated omni-modal models, even questions designed to require joint audio-visual reasoning derive most decision-relevant sensory information from modality-unique signals, while sensory synergy remains limited. This reveals a *sensory synergy bottleneck* in the evaluated systems.

4.1.2. FUNCTIONAL VALIDATION VIA MODALITY SHUFFLING

To examine whether this sensory imbalance is reflected in observable model behavior, we independently shuffle video or audio streams across samples. The intervention breaks the correspondence between the question and one sensory stream while preserving its modality-specific marginal distribution.

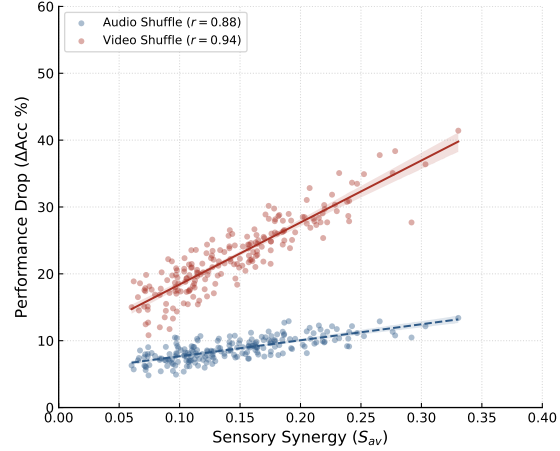


Figure 2. Functional validation via modality shuffling. Correlation between Sensory Synergy (S_{av}) and accuracy drop (ΔAcc) on the AV-Fusion subset of MUSIC-AVQA (Qwen2.5-Omni 7B) under audio or video shuffling. Each point represents a question instance; Spearman ρ is reported.

Consistent with the Sensory PID spectrum in Table 1, shuffling the video stream induces substantially larger performance drops than shuffling the audio stream, even on AV-Fusion questions. This asymmetric sensitivity indicates that model predictions depend more strongly on correctly aligned visual information than on auditory information under these modality-level perturbations.

Finding 2. Modality-shuffling interventions support a visually dominant decision profile: the evaluated omni-modal models suffer larger performance drops under visual than auditory misalignment, while Sensory PID is associated with sensitivity to cross-modal disruption.

4.2. Layer-Wise Sensory Dynamics

The final-layer sensory imbalance raises a further question: when does this asymmetry emerge during computation? We therefore trace Sensory PID across network depth to examine the temporal organization of video-, audio-, and jointly derived decision information.

4.2.1. VISUAL-FIRST DYNAMICS IN OMNI-MODAL MODELS

Figure 3(a)–(c) reveals a pronounced asymmetry in the temporal emergence of video-, audio-, and jointly derived information. Although the overall progression moves from weak early information toward stronger multimodal information in later layers, the two sensory streams do not become decision-relevant at comparable rates.

Early visual dominance. U_{vis} rises substantially earlier and more sharply than U_{aud} , becoming the dominant sensory component through much of the intermediate computation.

Table 1. Sensory PID spectrum and performance across Audio-Focused, Visual-Focused, and AV-Fusion subsets. U_{vis} , U_{aud} , S_{av} , and R_{sens} denote unimodal visual contribution, unimodal audio contribution, audiovisual synergy, and sensory redundancy, respectively, computed via conditional PID on $I(Y; V, A | T)$. The dominant information component for each model–subset pair is highlighted in **bold**. Acc and $Acc1$ report full audio-visual accuracy and visual-only accuracy under audio removing.

Model	Audio-Focused						Visual-Focused						AV-Fusion						Full Selection					
	U_{vis}	U_{aud}	S_{av}	R_{sens}	Acc	$Acc1$	U_{vis}	U_{aud}	S_{av}	R_{sens}	Acc	$Acc1$	U_{vis}	U_{aud}	S_{av}	R_{sens}	Acc	$Acc1$	U_{vis}	U_{aud}	S_{av}	R_{sens}	Acc	$Acc1$
VITA-1.5 7B	0.54	0.59	0.20	0.005	51.6	25.4	1.34	0.25	0.15	0.005	77.5	75.8	1.35	0.60	0.22	0.004	57.8	56.2	1.09	0.55	0.22	0.003	61.4	54.2
Qwen2.5-Omni 3B	0.48	0.64	0.15	0.004	48.0	22.0	1.26	0.22	0.12	0.004	70.0	68.0	1.25	0.55	0.21	0.003	52.0	50.0	1.06	0.50	0.18	0.001	55.0	48.0
Qwen2.5-Omni 7B	0.71	0.69	0.26	0.001	55.2	42.1	1.36	0.28	0.17	0.002	79.0	77.0	1.42	0.65	0.32	0.004	62.0	58.0	1.12	0.58	0.24	0.004	64.0	56.0

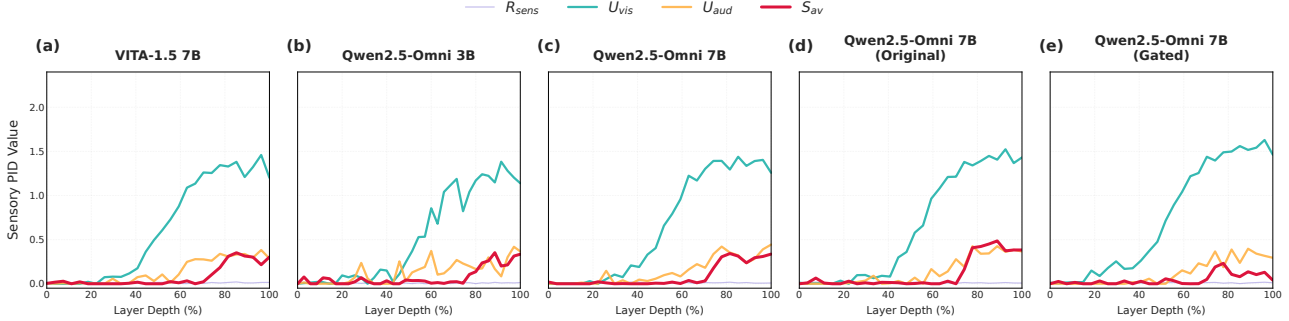


Figure 3. Layer-wise Sensory PID and instruction-conditioned fusion in omni-modal models. Sensory PID across depth for VITA-1.5 and Qwen2.5-Omni on the AV-Fusion subset of MUSIC-AVQA. Panels (a–c) plot U_{vis} , U_{aud} , R_{sens} , and S_{av} versus normalized depth. Panels (d–e) illustrate the effect of replacing a fusion-demanding instruction with a less fusion-dependent alternative while holding the sensory inputs fixed.

Auditory-unique information remains comparatively limited during this stage. This ordering indicates that visual evidence becomes decision-relevant earlier, producing an already visually skewed predictive state before substantial auditory or cross-sensory information emerges.

Constrained late sensory integration. Sensory synergy S_{av} increases primarily in later layers, indicating that audiovisual integration is delayed relative to the accumulation of visual-unique information. Although this late-stage rise indicates increasing joint use of the two sensory streams, its magnitude generally remains below U_{vis} . Together with the final-layer decomposition in Table 1, these dynamics suggest that the sensory synergy bottleneck is not simply a final-output phenomenon: it is associated with a temporal imbalance in which visual information becomes decision-relevant earlier than auditory and synergistic information.

Architecture and scale modulate timing, but not the ordering. The precise trajectories vary across model families. VITA-1.5 exhibits an earlier and steeper increase in U_{vis} , whereas Qwen2.5-Omni shows a more gradual rise before converging to a similarly visual-dominant profile in later layers. Model scale also affects the smoothness and magnitude of these trajectories. Across the evaluated architectures and scales, however, the qualitative ordering remains consistent: visual-unique information emerges first, while auditory-unique information and audiovisual synergy

become prominent later.

Finding 3. Across the evaluated omni-modal models, unique visual information emerges earlier than both unique auditory information and audiovisual synergy. This consistent temporal ordering reveals a *visual-first computation pattern*: intermediate decisions become visually dominated before stronger cross-sensory integration develops in later layers.

4.2.2. INSTRUCTION-CONDITIONED SENSORY FUSION

We finally examine whether the late-stage audiovisual synergy observed above varies with the linguistic task specification. Holding the video and audio streams fixed, we modify the instruction to reduce its demand for joint audiovisual reasoning and re-estimate the layer-wise Sensory Synergy S_{av} .

Figure 3(d)–(e) provides an illustrative example. Replacing a fusion-demanding query such as “*Is the violin playing the high pitch?*” with a less fusion-dependent alternative such as “*Which instrument is being played?*” substantially attenuates late-layer S_{av} , while leaving the early- and mid-layer unimodal trajectories comparatively stable. Thus, identical sensory inputs can exhibit different integration profiles depending on the task specified by the language instruction.

This observation complements the visual-first pattern above: sensory evidence begins to accumulate before substantial

audiovisual synergy emerges, while the extent of late-stage fusion varies with the instructional demand for joint sensory reasoning. Language therefore serves as a task-conditioning context for the sensory interaction analyzed by Sensory PID.

Finding 4. Holding video and audio fixed, changing the instruction alters late-stage audiovisual synergy while leaving earlier unimodal dynamics comparatively stable. This provides evidence for *instruction-conditioned sensory fusion*: the extent to which joint sensory evidence becomes decision-relevant varies with the linguistic task demand.

5. Evaluating PID-Guided Reweighting

We now evaluate the PID-guided sample-reweighting policy introduced in Section 2.3. These experiments serve as a separate vision–language proof of concept for testing whether local information-decomposition diagnostics provide useful training guidance beyond uniform fine-tuning, generic sample difficulty, or simple modality sensitivity. We evaluate both downstream performance and the post-tuning PID profile to examine where the proposed policy is effective and where its benefits are limited.

5.1. Experimental Setup and Main Results

Experimental setup. We profile 3,000 MMBench training samples and compare LoRA-PID against standard uniform LoRA and three matched-count reweighting controls: random assignment, accuracy-based difficulty mining, and modality-ablation-based selection. The primary experiment uses Qwen2.5-VL-7B, with additional transfer experiments on LLaVA-OneVision-7B and Gemma3-12B. Together, these controls test whether the PID-guided policy provides information beyond non-uniform weighting, generic sample difficulty, or simple sensitivity to visual input.

All reweighting conditions use identical sample counts and LoRA hyperparameters. Adapters are placed in the final 20% of transformer layers. No evaluation samples are used for PID profiling, hyperparameter selection, or LoRA training. Results are reported over three random seeds, and post-tuning PID diagnostics are estimated on a held-out MMStar subset. Full experimental conditions, data controls, and LoRA configurations are provided in Appendix H.

PID-guided reweighting outperforms matched alternatives. Table 2 shows that LoRA-PID improves all three reasoning- and grounding-oriented benchmarks relative to uniform fine-tuning: +1.1 pp on MMBench, +2.3 pp on MMStar, and +1.3 pp on POPE. The MMStar improvement over uniform LoRA is supported by a 95% paired-bootstrap confidence interval of [1.4, 3.2] pp. The MMStar

advantage also remains positive relative to random reweighting (+2.8 pp, 95% CI [1.6, 4.0]), accuracy-based mining (+1.8 pp, [0.7, 2.9]), and ablation-based selection (+1.3 pp, [0.2, 2.4]). Full statistics are reported in Appendix H.4.

On MMStar, these matched comparisons indicate that the observed gain is not explained solely by assigning non-uniform weights, selecting generically difficult samples, or prioritizing samples that are simply sensitive to the addition of vision. Instead, the structured combination of fusion potential, current synergy, and text-unique reliance provides a more informative ranking signal in the evaluated reasoning-oriented setting.

The trade-off is selective. Relative to uniform LoRA, LoRA-PID decreases MMMU by 0.5 pp and PMC-VQA by 0.3 pp. This trade-off is consistent with the policy’s downweighting of examples exhibiting high text-unique reliance and suggests that the intervention preferentially benefits settings in which stronger joint-modality use is useful.

5.2. Where Does PID-Guided Reweighting Help?

Gains concentrate on reasoning-intensive categories. The aggregate improvement is not uniform across MM-Bench categories. Relative to uniform LoRA, PID-guided reweighting improves Spatial Reasoning by +3.5 pp, Attribute Comparison by +2.8 pp, and Action Recognition by +2.5 pp. The gains are smaller for Object Localization (+1.1 pp) and Scene Classification (+0.5 pp), and are slightly negative for OCR/Text Recognition and Commonsense (−0.3 pp each).

This pattern is consistent with the intended role of the reweighting policy: the largest gains occur in categories labeled as reasoning-oriented in our analysis, whereas the improvements are smaller or slightly negative for the evaluated perception- and knowledge-oriented categories. The complete category-level results are reported in Appendix H.4.

A similar pattern appears on POPE: the largest gain over uniform LoRA occurs on the Adversarial subset (+1.9 pp), where shortcut-prone co-occurrence cues are particularly challenging (Appendix H.4).

The post-tuning information profile shifts in the targeted direction. LoRA-PID produces the largest increase in held-out MMStar synergy, from $S_{vl} = 1.20$ after uniform LoRA to $S_{vl} = 1.36$, while U_{txt} decreases from 0.56 to 0.46. Consequently, the synergy share increases from 67.5% to 73.9%. The matched alternative reweighting methods produce smaller shifts in the corresponding post-tuning PID profile.

Thus, the downstream improvement is accompanied by a change in the model-induced information profile in the direc-

Table 2. **PID-guided reweighting results on Qwen2.5-VL-7B.** Values are mean $\pm$ std over three seeds. Post- S_{vl} and Post- U_{txt} are estimated on a held-out MMStar subset.

Condition	Reasoning		Grounding	Prior-Dominated		PID Diagnostics	
	MMBench	MMStar	POPE	MMMU	PMC-VQA	Post- S_{vl}	Post- U_{txt}
A: Base	88.3	60.7	86.4	54.1	53.1	1.15	0.60
B: LoRA-Uniform	89.1 $\pm$.3	62.0 $\pm$.4	87.2 $\pm$.3	54.0 $\pm$.3	53.0 $\pm$.3	1.20 $\pm$.02	0.56 $\pm$.02
C: LoRA-PID	90.2$\pm$.2	64.3$\pm$.3	88.5$\pm$.2	53.5 $\pm$.4	52.7 $\pm$.3	1.36$\pm$.03	0.46$\pm$.02
D: LoRA-Random	88.8 $\pm$.5	61.5 $\pm$.6	86.9 $\pm$.4	54.1 $\pm$.4	53.2 $\pm$.4	1.16 $\pm$.04	0.58 $\pm$.03
E: LoRA-Acc	89.4 $\pm$.3	62.5 $\pm$.4	87.5 $\pm$.3	54.3$\pm$.3	53.4$\pm$.3	1.22 $\pm$.02	0.54 $\pm$.02
F: LoRA-Ablation	89.6 $\pm$.3	63.0 $\pm$.4	87.8 $\pm$.3	53.8 $\pm$.3	52.9 $\pm$.3	1.24 $\pm$.02	0.52 $\pm$.02

Table 3. **Cross-model behavior of PID-guided reweighting.** Values compare uniform LoRA and PID-guided LoRA. Post- S_{vl} is measured on held-out MMStar.

Model	Condition	MMBench	MMStar	Post- S_{vl}
LLaVA-OV-7B	LoRA-Uniform	85.5 $\pm$.3	60.0 $\pm$.4	1.00 $\pm$.02
	LoRA-PID	86.3$\pm$.3	61.5$\pm$.4	1.10$\pm$.03
Gemma3-12B	LoRA-Uniform	83.1 $\pm$.4	56.2$\pm$.5	0.20 $\pm$.02
	LoRA-PID	83.3$\pm$.4	54.3 $\pm$.5	0.21 $\pm$.02

tion targeted by the reweighting policy. This correspondence does not by itself establish a causal relationship between increased synergy and performance, but it provides evidence that the intervention changes the diagnostic profile it was designed to target. The full decomposition is reported in Appendix H.4.

5.3. Cross-Model Behavior and Boundary Case

To examine whether the intervention transfers across different modality-use profiles, we repeat the comparison on LLaVA-OneVision-7B and Gemma3-12B. Table 3 reports the uniform-LoRA and PID-guided conditions; complete results are provided in Appendix H.5.

For LLaVA-OneVision-7B, PID-guided reweighting improves MMBench from 85.5 to 86.3 and MMStar from 60.0 to 61.5, while increasing post-tuning S_{vl} from 1.00 to 1.10. This reproduces the same qualitative direction observed for Qwen2.5-VL on a second model with meaningful joint-modality use.

In contrast, the strongly language-prior-reliant Gemma3-12B does not show the same benefit. Its MMBench score changes only from 83.1 to 83.3, post-tuning S_{vl} changes from 0.20 to 0.21, and MMStar decreases from 56.2 to 54.3. The complete results in Appendix H.5 further show that MMMU decreases from 59.3 after uniform LoRA to 54.1 after PID-guided reweighting.

This contrast supports a profile-dependent interpretation: PID-guided reweighting appears most useful when the pre-trained model already exhibits meaningful joint-modality use that can be amplified through data reweighting. The present experiments do not establish that the policy can create such fusion capacity when the model’s baseline decisions remain strongly language-prior-reliant.

Finding 5. PID-guided sample reweighting provides an actionable but profile-dependent intervention. In the evaluated joint-modality-reliant models, it improves the reported reasoning and grounding metrics while shifting post-tuning decisions toward higher cross-modal synergy; for Qwen2.5-VL-7B, the MMStar gain is additionally supported by paired-bootstrap confidence intervals. The same benefit is not observed for the evaluated strongly language-prior-reliant Gemma3-12B.

6. Related Work

The shift from vision–language to omni-modal models marks a transition in generative AI (Luo et al., 2025; Chen et al., 2024a; Xu et al., 2025; Guo et al., 2025). Existing analyses based on attention, representation alignment, and modality ablation provide complementary views of multi-modal behavior (Wang & Wang, 2025; Basile et al., 2025; Fang et al., 2026). PID has been used to characterize multimodal information structure (Fang et al., 2026), and Xiu et al. (2026) further introduced decision-level PID analysis for modality interaction in LVLMs. However, how omni-modal models integrate multiple sensory streams under language-conditioned settings remains unclear. We address this question through *Sensory PID*, which decomposes video–audio information into unique, redundant, and synergistic components. Local PID diagnostics further provide signals for VLM sample reweighting. See Appendix B.

7. Conclusion

We introduce *Sensory PID*, a conditional decision-level decomposition for language-conditioned video–audio interaction in omni-modal models. It reveals a sensory synergy bottleneck, visual dominance, visual-first dynamics, and instruction-conditioned sensory fusion. Local PID diagnostics further provide actionable signals for VLM sample reweighting, improving reasoning and grounding while increasing post-tuning cross-modal synergy. Analogous vision–language analyses serve as reference validation and broadly recover prior decision-level PID patterns. See limitations in Appendix C.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Methodology Details and Derivations

This appendix provides methodological details complementary to the omni-modal Sensory PID formulation in the main text. We first describe the analogous two-source vision–language PID formulation used for the reference analyses and PID-guided reweighting experiments. We then provide the underlying BATCH estimation procedure, the practical vision–language implementation details, and the derivation of the BATCH-induced local diagnostic scores used in §5.

A.1. Vision–Language PID Reference

For the vision–language reference setting, let Y denote the model’s prediction over a finite candidate set $\mathcal{C}$ (*e.g.*, multiple-choice options: A, B, C, D), and let X_v and X_t denote the visual and textual source variables, respectively. We construct an empirical decision distribution by pairing the data distribution with the model-induced predictive distribution:

$$P(x_v, x_t, y) = P_{\mathcal{D}}(x_v, x_t) p_{\theta}(y \mid x_v, x_t), \quad (13)$$

with corresponding partial-input distributions induced by $p_{\theta}(y \mid x_v)$ and $p_{\theta}(y \mid x_t)$ through the calibrated embedding-masking procedure described in Appendix A.2. All information quantities below are therefore computed with respect to model-induced predictive distributions rather than ground-truth labels or latent representations.

Partial Information Decomposition (PID) (Williams & Beer, 2010) partitions the decision-relevant information provided by the two sources into four non-negative components:

$$I(Y; X_v) = R_{v1} + U_{\text{vis}}, \quad (14)$$

$$I(Y; X_t) = R_{v1} + U_{\text{txt}}, \quad (15)$$

$$I(Y; X_v, X_t) = R_{v1} + U_{\text{vis}} + U_{\text{txt}} + S_{v1}, \quad (16)$$

where U_{vis} and U_{txt} denote information uniquely attributable to vision and text, respectively; R_{v1} denotes information redundantly available from either source; and S_{v1} denotes information available only from their joint observation.

Following Bertschinger et al. (2014), these components are defined through optimization over the set Δ_P of joint distributions $Q(X_v, X_t, Y)$ that preserve the source–target marginals:

$$\Delta_P = \{Q : Q(x_v, y) = P(x_v, y), Q(x_t, y) = P(x_t, y)\}. \quad (17)$$

Let

$$M_{\min} = \min_{Q \in \Delta_P} I_Q(Y; X_v, X_t). \quad (18)$$

The synergistic component is defined as the gap between the true joint mutual information and the minimum achievable under the preserved source–target marginals:

$$S_{v1} = I_P(Y; X_v, X_t) - M_{\min}. \quad (19)$$

The remaining components follow algebraically:

$$U_{\text{txt}} = M_{\min} - I_P(Y; X_v), \quad (20)$$

$$U_{\text{vis}} = M_{\min} - I_P(Y; X_t), \quad (21)$$

$$R_{v1} = I_P(Y; X_v) - U_{\text{vis}} = I_P(Y; X_t) - U_{\text{txt}}. \quad (22)$$

The four components are non-negative and sum to $I_P(Y; X_v, X_t)$ by construction.

We use this decomposition as an analogous two-source reference for the vision–language analyses and for the local PID diagnostics employed in §5. Our methodological focus is instead the conditional Sensory PID formulation $I(Y; V, A \mid T)$ for language-conditioned video–audio interaction. The decision-level vision–language formulation follows the framework used in prior LVLM PID analysis (Xiu et al., 2026) and is not treated as a methodological contribution of this work.

A.2. Scalable Estimation via BATCH

The PID definitions above require optimization over $Q \in \Delta_P$, which is intractable for the high-dimensional, continuous representations produced by modern MLLMs. We use the BATCH estimator (Liang et al., 2023) as the underlying numerical

estimator. BATCH amortizes the constrained optimization through learned source encoders and Sinkhorn-normalized couplings.

The derivation below is written using the vision–language notation (X_v, X_t) because the same two-source formulation underlies the reference analyses and the local diagnostic scores used for VLM reweighting in §5. In the omni-modal analysis, BATCH is analogously applied to the video and audio sensory sources while the textual instruction is retained as conditioning context.

Neural parameterization. BATCH learns a parametric distribution $\tilde{Q}_\phi$ via two encoder networks f_{ϕ_1} and f_{ϕ_2} that map X_v and X_t into a shared latent space. The unnormalized affinity between sample pairs within a mini-batch $\mathcal{B}$ of size m is

$$A_{ij}^{(y)} = \exp\left(f_{\phi_1}(x_v^{(i)}, y)^\top f_{\phi_2}(x_t^{(j)}, y)\right), \quad i, j \in \{1, \dots, m\}. \quad (23)$$

Marginal constraints via Sinkhorn–Knopp. The PID definition requires $Q \in \Delta_P$. Within a mini-batch, this corresponds to preserving the required source marginals. BATCH enforces the constraints by applying the Sinkhorn–Knopp algorithm (Cuturi, 2013) to $A^{(y)}$ for each label y , iteratively scaling rows and columns until the resulting matrix $\tilde{Q}^{(y)}$ is doubly stochastic.

Optimization objective. The estimator minimizes an upper bound on $I_{\tilde{Q}}(Y; X_v, X_t)$ via gradient descent on the encoder parameters:

$$\mathcal{L}(\phi) = \mathbb{E}_{\tilde{Q}} \left[\log \frac{\tilde{Q}(x_t | x_v, y) \tilde{Q}(x_v | y)}{\sum_{y'} \tilde{Q}(x_t | x_v, y') \tilde{Q}(y' | x_v) \tilde{Q}(x_v)} \right]. \quad (24)$$

PID term recovery. Upon convergence, the four PID terms are computed by comparing the data distribution P against the optimized $\tilde{Q}$:

$$S_{vl} = I_P(Y; X_v, X_t) - I_{\tilde{Q}}(Y; X_v, X_t), \quad (25)$$

$$U_{\text{txt}} = I_{\tilde{Q}}(Y; X_v, X_t) - I_P(Y; X_v), \quad (26)$$

$$U_{\text{vis}} = I_{\tilde{Q}}(Y; X_v, X_t) - I_P(Y; X_t), \quad (27)$$

$$R_{vl} = I_{\tilde{Q}}(Y; X_v, X_t) - U_{\text{txt}} - U_{\text{vis}}. \quad (28)$$

Vision–language source representations. Token-based MLLMs produce variable-length modality sequences whereas BATCH requires fixed-dimensional continuous source variables. For the vision–language reference analysis, we therefore summarize each modality using mean pooling (Shen et al., 2018) after the modality tokens have been mapped or embedded into the shared multimodal token space. This yields source representations $X_v, X_t \in \mathbb{R}^d$. The parameter-free aggregation provides a global, sample-aligned modality summary for information estimation. Sensitivity to alternative feature summarization strategies is reported in Appendix G.

Calibrated vision–language masking. Jointly trained MLLMs do not always expose native unimodal prediction paths. We therefore approximate the partial-input predictive distributions through representation corruption, following the broader use of corruption-based interventions (Meng et al., 2022). To estimate $p_\theta(y | x_v)$, we replace the projected text embeddings while retaining the visual input; conversely, to estimate $p_\theta(y | x_t)$, we replace the visual embeddings while retaining the textual input. For a masked modality m ,

$$\tilde{E}_m[\ell] \sim \mathcal{N}(\mu_m, \text{diag}(\sigma_m^2)), \quad \ell = 1, \dots, L_m, \quad (29)$$

where μ_m and σ_m are per-dimension statistics estimated over the profiling set $\mathcal{D}_{\text{prof}}$. Matching these empirical statistics controls the perturbation scale while removing instance-specific information from the masked modality.

Output stabilization. For finite-candidate prediction, we apply three stabilization steps before PID estimation. First, *valid-option renormalization* restricts predictions to the candidate set $\mathcal{C}$ through selective softmax over the corresponding logits. Second, *confidence gating* prevents low-confidence predictions from becoming artificially sharp after renormalization: when the total probability mass assigned to valid options falls below $\tau = 0.3$, we replace the output with a uniform

distribution over $\mathcal{C}$. Third, rather than constructing the target marginal from hard argmax labels, we preserve predictive uncertainty through *soft aggregation*:

$$\hat{p}(y) = \frac{1}{|\mathcal{D}_{\text{prof}}|} \sum_{j \in \mathcal{D}_{\text{prof}}} p_{\theta}(y \mid x_v^{(j)}, x_t^{(j)}), \quad (30)$$

and keep this marginal fixed during BATCH optimization.

Estimator provenance. BATCH, mean pooling, embedding-space corruption, confidence regularization, and soft aggregation are estimation machinery rather than methodological contributions of this work. Closely related combinations of these choices have been reported in prior decision-level LVLM analysis (Xiu et al., 2026). The methodological contribution emphasized in the main text is instead the conditional Sensory PID formulation and its instantiation for language-conditioned video–audio interaction.

A.3. BATCH-Induced Local Diagnostic Scores

For the PID-guided reweighting application in §5, we further retain the sample-level additive contributions underlying the aggregate vision–language PID estimates. The four mutual-information quantities underlying the decomposition— $I_P(Y; X_v)$, $I_P(Y; X_t)$, $I_P(Y; X_v, X_t)$, and $I_{\tilde{Q}}(Y; X_v, X_t)$ —are implemented as expectations over samples. Retaining their sample-level terms before averaging yields the BATCH-induced local diagnostic contributions

$$s_i = \text{mi}_P^{(i)}(Y; X_v, X_t) - \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t), \quad (31)$$

$$u_{\text{txt},i} = \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t) - \text{mi}_P^{(i)}(Y; X_v), \quad (32)$$

$$u_{\text{vis},i} = \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t) - \text{mi}_P^{(i)}(Y; X_t), \quad (33)$$

$$r_i = \text{mi}_P^{(i)}(Y; X_v) + \text{mi}_P^{(i)}(Y; X_t) - \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t). \quad (34)$$

For example,

$$\text{mi}_P^{(i)}(Y; X_v) = \sum_y p_{\theta}(y \mid x_v^{(i)}) \log \frac{p_{\theta}(y \mid x_v^{(i)})}{\hat{p}(y)}, \quad (35)$$

is the local MI contribution from the vision-only predictive distribution, while $\text{mi}_{\tilde{Q}}^{(i)}$ additionally involves the learned Sinkhorn coupling.

Within each batch context, the aggregate PID estimate is recovered by averaging these local contributions: $\hat{S} = |\mathcal{B}|^{-1} \sum_i s_i$, and analogously for $\hat{U}_{\text{txt}}$, $\hat{U}_{\text{vis}}$, and $\hat{R}$.

These local scores are *not* axiomatic pointwise PID atoms—a formal theory of pointwise PID remains an open problem (Liang et al., 2023). We use them exclusively as BATCH-induced additive *ranking signals*. Because the Sinkhorn coupling depends on batch composition, we stabilize each sample’s score by averaging over $K = 50$ random batch draws with batch size 256.

To construct the reweighting signals used in §5, we apply non-negative clipping to the local contributions and define the per-sample information mass

$$I_i^+ = [s_i]_+ + [u_{\text{vis},i}]_+ + [u_{\text{txt},i}]_+ + [r_i]_+, \quad [z]_+ = \max(z, 0). \quad (36)$$

The *synergy ratio* and *shortcut score* are then

$$\text{SR}_i = \frac{[s_i]_+}{I_i^+ + \epsilon}, \quad \text{SC}_i = \frac{[u_{\text{txt},i}]_+}{I_i^+ + \epsilon}, \quad (37)$$

where $\epsilon = 10^{-8}$. SR_i serves as a proxy for the relative synergy usage of sample i , while SC_i captures its relative text-unique reliance. These ratios convert aggregate interaction diagnostics into sample-level ranking signals used by the PID-guided reweighting policy in §5.

A.4. BATCH-Induced Local Diagnostic Scores: Full Derivation

Appendix A.3 introduces the BATCH-induced sample-level diagnostic scores used for reweighting in §5. Here we provide the complete derivation and the stability checks used in our experiments.

Per-sample MI terms. The four MI quantities underlying the PID— $I_P(Y; X_v)$, $I_P(Y; X_t)$, $I_P(Y; X_v, X_t)$, and $I_{\tilde{Q}}(Y; X_v, X_t)$ — are each implemented as sample-level expectations. The partial-input terms use the calibrated vision-language masking procedure described in Appendix A.2:

$$\text{mi}_P^{(i)}(Y; X_v) = \sum_{y \in \mathcal{C}} p_\theta(y | x_v^{(i)}) \log \frac{p_\theta(y | x_v^{(i)})}{\hat{p}(y)}, \quad (38)$$

$$\text{mi}_P^{(i)}(Y; X_t) = \sum_{y \in \mathcal{C}} p_\theta(y | x_t^{(i)}) \log \frac{p_\theta(y | x_t^{(i)})}{\hat{p}(y)}, \quad (39)$$

$$\text{mi}_P^{(i)}(Y; X_v, X_t) = \sum_{y \in \mathcal{C}} p_\theta(y | x_v^{(i)}, x_t^{(i)}) \log \frac{p_\theta(y | x_v^{(i)}, x_t^{(i)})}{\hat{p}(y)}. \quad (40)$$

Here

$$\hat{p}(y) = \frac{1}{|\mathcal{D}_{\text{prof}}|} \sum_j p_\theta(y | x_v^{(j)}, x_t^{(j)}) \quad (41)$$

is the marginal label distribution estimated once from the profiling set.

Row-anchored coupling MI. The fourth term additionally involves the learned Sinkhorn coupling. Fixing the X_v index at sample i and marginalizing over all X_t samples in the batch gives

$$\begin{aligned} \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t) &= \sum_{j \in \mathcal{B}} \sum_y p_\theta(y | x_v^{(i)}) \tilde{q}(x_t^{(j)} | x_v^{(i)}, y) \\ &\quad \times \log \frac{p_\theta(y | x_v^{(i)}) \tilde{q}(x_t^{(j)} | x_v^{(i)}, y)}{\hat{p}(y) \tilde{q}(x_t^{(j)} | x_v^{(i)})}. \end{aligned} \quad (42)$$

Here $\tilde{q}(x_t^{(j)} | x_v^{(i)}, y)$ is obtained by normalizing the i -th row of the Sinkhorn matrix associated with label y , and

$$\tilde{q}(x_t^{(j)} | x_v^{(i)}) = \sum_{y'} \tilde{q}(x_t^{(j)} | x_v^{(i)}, y') p_\theta(y' | x_v^{(i)}). \quad (43)$$

Local PID scores. Substituting the sample-level MI quantities above into the aggregate PID identities gives

$$s_i(\mathcal{B}) = \text{mi}_P^{(i)}(Y; X_v, X_t) - \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t), \quad (44)$$

$$u_{\text{txt}, i}(\mathcal{B}) = \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t) - \text{mi}_P^{(i)}(Y; X_v), \quad (45)$$

$$u_{\text{vis}, i}(\mathcal{B}) = \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t) - \text{mi}_P^{(i)}(Y; X_t), \quad (46)$$

$$r_i(\mathcal{B}) = \text{mi}_P^{(i)}(Y; X_v) + \text{mi}_P^{(i)}(Y; X_t) - \text{mi}_{\tilde{Q}}^{(i)}(Y; X_v, X_t). \quad (47)$$

The batch-level PID estimates recover by averaging these terms:

$$\hat{S} = \frac{1}{|\mathcal{B}|} \sum_i s_i(\mathcal{B}), \quad (48)$$

and analogously for $\hat{U}_{\text{txt}}$, $\hat{U}_{\text{vis}}$, and $\hat{R}$.

Row vs. column anchoring. One can alternatively construct the local coupling contribution by fixing X_t and marginalizing over X_v , yielding a column-anchored score. We use row-anchored scores by default and evaluate whether this choice materially changes sample ranking. Across the 3,000 profiled samples, the Spearman correlation between row- and column-anchored $\bar{s}_i$ scores is $\rho = 0.87$ (95% CI [0.83, 0.91]), indicating broadly consistent rankings.

Multi-context averaging. Because the Sinkhorn coupling depends on the composition of the mini-batch, an individual sample’s local score can vary across batch contexts. We therefore average each score over $K = 50$ random batch draws with batch size 256:

$$\bar{s}_i = \frac{1}{K} \sum_{k=1}^K s_i(\mathcal{B}_k), \quad \bar{u}_{\text{txt},i} = \frac{1}{K} \sum_{k=1}^K u_{\text{txt},i}(\mathcal{B}_k). \quad (49)$$

The same averaging procedure is applied to the remaining local diagnostic contributions before constructing the reweighting scores used in §5.

B. Extended Related Work

Multimodal Large Language Models and Evaluation. The progression from vision–language models (VLMs) to omni-modal systems capable of processing interleaved video, audio, and text marks a major shift in generative AI (Fu et al., 2025; Xu et al., 2025). Early systems typically relied on separate modality encoders connected to a language-model backbone, while recent native multimodal architectures increasingly fuse sensory inputs within a shared token space (Zhang et al., 2026a; Dai et al., 2023; Liu et al., 2024a; Luo et al., 2025). Despite strong benchmark performance, evaluation remains largely outcome-oriented, relying on accuracy-based leaderboards (Zheng et al., 2023) or human preference ratings (Zhang et al., 2025a). Such metrics are useful for ranking models, but they provide limited insight into how different modalities contribute to the final decision. Our work shifts the focus from *what* a model predicts to *how* sensory and linguistic information is used at the decision level.

Interpretability and Modality Interaction Analysis. Recent work on LLM interpretability has developed a broad set of tools for explaining model behavior beyond final predictions, including attribution, probing, mechanistic analysis, and representation-level inspection (Ma et al., 2026; Zhang et al., 2025b; 2026b). Understanding modality interaction is essential for reliable multimodal deployment. Existing approaches can be broadly grouped into attention analysis, representation alignment, and modality ablation. Attention-based methods visualize token- or head-level associations (Jain & Wallace, 2019; Chefer et al., 2021; Basile et al., 2025), but attention weights do not necessarily reflect functional contribution. Representation analyses measure geometric or statistical alignment between modality embeddings (Morcos et al., 2018; Nguyen et al., 2020; Radford et al., 2021); however, high alignment does not imply that the aligned information is used in the final prediction. Ablation studies quantify behavioral sensitivity under modality removal or perturbation (Jiao et al., 2024; Tong et al., 2024; Thrush et al., 2022), but they conflate qualitatively different sources of dependence: a performance drop may arise from modality-unique information, redundant evidence, or genuinely synergistic integration. In contrast, our PID framework explicitly separates unique, redundant, and synergistic decision information, enabling a finer-grained characterization of modality use than accuracy, alignment, or ablation alone.

Information-Theoretic Analysis in Deep Learning. Information theory provides a general framework for quantifying uncertainty, dependence, and information flow, and has been widely applied across communication, neuroscience, complex systems, and machine learning (Du et al., 2024; Kim et al., 2026; Huang et al., 2026). The Information Bottleneck principle (Tishby & Zaslavsky, 2015) has been widely used to study compression and prediction in neural networks, while Partial Information Decomposition (PID) (Williams & Beer, 2010) separates the information that multiple sources provide about a target into unique, redundant, and synergistic components. Recent estimators and scalable approximations (Faes et al., 2016; Kolchinsky, 2022; Wibral et al., 2017; Bertschinger et al., 2014; Liang et al., 2023) have made PID increasingly applicable to high-dimensional systems. In multimodal learning, PID has been used to characterize redundancy–uniqueness–synergy structure across modality pairs and to inform representation alignment (Fang et al., 2026). More recently, Xiu et al. (2026) study PID over model-induced predictive distributions in LVLMs, enabling decision-level analysis of vision–language interaction; related information-decomposition perspectives have also been explored in diffusion models (Dewan et al., 2024).

Our work focuses on a complementary setting in which language specifies the task and video and audio provide sensory evidence. We introduce *Sensory PID*, which conditions on language and decomposes $I(Y; V, A \mid T)$ into video-unique, audio-unique, redundant, and audiovisual-synergistic information. This formulation supports analyses specific to omni-modal

sensory interaction, including modality-specific perturbations, visual-first layer-wise dynamics, and instruction-conditioned sensory fusion. Separately, we use BATCH-induced local PID diagnostics as ranking signals for VLM sample reweighting, connecting information-theoretic diagnosis with a proof-of-concept training intervention.

C. Limitations and Future Directions

While PID and Sensory PID provide a principled framework for analyzing multimodal interaction, we acknowledge several limitations of the current study.

Constrained Decoding Space. First, our estimation framework operates over a discrete decision space, primarily multiple-choice QA. This setting makes the model’s predictive distribution tractable and enables stable PID estimation, but it does not fully capture open-ended generation. Extending PID to free-form responses remains challenging because mutual information must be estimated over high-dimensional, sequential output distributions. Future work will require more efficient estimators for sequence-level or semantic-level decision spaces.

Approximation of Unimodal Conditionals. Second, our training-free analysis relies on calibrated embedding masking to approximate unimodal conditionals such as $P(Y \mid V, T)$ or $P(Y \mid A, T)$. Although moment-matched noise helps preserve distributional stability, masking remains a proxy for native unimodal inference. The resulting hidden states may differ from those produced by models explicitly trained or prompted for partial-modality inputs, potentially introducing estimation bias in unique information terms such as U_{vis} and U_{aud} .

Statistical Rather than Causal Interpretation. Third, PID measures statistical dependence in model-induced predictive distributions; it is not a causal calculus. While PID profiles predict sensitivity to modality-level interventions, the PID terms themselves do not identify the internal circuits, attention heads, or neurons responsible for the observed behavior. Thus, high synergy is consistent with functional cross-modal integration, but it does not by itself reveal the mechanistic implementation of that integration.

Scope of PID-Guided Reweighting. Finally, our PID-guided reweighting experiment is intended as a proof-of-concept rather than a complete training recipe. We use lightweight LoRA fine-tuning and a limited reweighting policy to test whether local diagnostic scores contain actionable signal. Scaling this idea to larger corpora, stronger adaptation methods, and open-ended instruction tuning remains an important direction for future work.

Future Directions. Building on these limitations, several directions are particularly promising.

- **Extending PID beyond multiple-choice decisions.** Future work could generalize the estimator to continuous or autoregressive output spaces, enabling analysis of chat-style assistants, dense captioning, and long-form multimodal generation.
- **Using PID as a training-time diagnostic.** PID terms could serve as monitoring signals during multimodal training or scaling analysis, helping identify when models rely on language shortcuts, redundant evidence, or genuine cross-modal integration. More speculative extensions could explore PID-inspired auxiliary objectives that encourage fusion-sensitive behavior, while carefully avoiding over-optimization of any single information term.
- **PID-guided benchmark and data construction.** PID analysis can help curate evaluation or training sets that better isolate cross-modal reasoning. For example, samples with high text-unique reliance may reveal language-prior shortcuts, whereas high-synergy samples can support more targeted evaluation of multimodal fusion. This may lead to more informative benchmarks than those defined solely by surface-level task categories.

D. Experimental Setup Details

D.1. Prompt Templates

To ensure fair evaluation and reproducible results, we employ a unified prompt structure across all benchmarks. Table 4 details the specific templates used for standard 4-option multiple-choice tasks (e.g., MMBench, MMMU) and binary 2-option tasks (e.g., specific subsets of MUSIC-AVQA or Boolean questions).

Table 4. Prompt templates used for zero-shot evaluation. The content within `< >` is replaced by the specific instance data during inference.

Task Type	Prompt Template
4-Option Questions (Standard MCQ)	Question: <code><Question Body></code> Options: (A) <code><Option A></code> (B) <code><Option B></code> (C) <code><Option C></code> (D) <code><Option D></code> Instruction: Please analyze and select the most appropriate answer. Output only the single letter corresponding to the correct option above (e.g., A, B, C, or D). Do not provide any explanation.
2-Option Questions (Binary/True-False)	Question: <code><Question Body></code> Options: (A) <code><Option A></code> (B) <code><Option B></code> Instruction: Please analyze and select the most appropriate answer. Output only the single letter corresponding to the correct option above (e.g., A or B). Do not provide any explanation.

D.2. Inference Settings

For all evaluations, we use deterministic decoding strategies to minimize variance. Specifically, we set the temperature to 0 and top-p to 1.0. The maximum output token length is restricted to ensure the model adheres to the single-letter constraint.

D.3. Hyperparameters and Training Details

To ensure the full reproducibility of our PID or sensory PID estimation results, we provide the detailed hyperparameter configurations used for the BATCH estimator. Following the guidelines established in previous work (Liang et al., 2023), we employed a lightweight Multi-Layer Perceptron (MLP) parameterization to approximate the marginal-matching joint distributions.

We adhered to a fixed configuration across all experiments (Vision-Language and Omni-modal tasks) to ensure consistency. Table 5 summarizes the specific settings. While the estimator is generally robust to hyperparameter variations, we report the exact values used to generate the main results in this paper.

E. Dataset Details and Taxonomy

In this section, we provide detailed specifications of the benchmarks used in our analysis. For the vision-language benchmarks, we organize datasets according to the **interaction regimes identified by our PID analysis** in the main text. These categories are analytical summaries of the observed model-benchmark profiles, rather than labels assigned before applying PID. We also describe the subsets constructed for the omni-modal analysis.

E.1. Vision-Language Benchmarks

Regime I: Synergy-Dominant Reasoning and Grounding Benchmarks. These benchmarks exhibit relatively high cross-modal synergy ($S_{v,l}$) in our experiments, suggesting that successful model decisions tend to depend on joint use of visual evidence and textual queries.

- **MMBench** (Liu et al., 2024b): MMBench is a systematically designed benchmark for evaluating LVLMs across approximately 20 fine-grained ability dimensions, ranging from basic object perception and localization to relation reasoning and attribute understanding. Its hierarchical taxonomy and quality-control pipeline make it a broad testbed

Table 5. Hyperparameters and configuration details for the BATCH Estimator.

Hyperparameter	Value / Configuration
<i>Optimization</i>	
Optimizer	Adam
Learning Rate Set	1×10^{-3} and 1×10^{-4}
Training Epochs Set	10, 15, 20
Batch Size Set (Training & Inference)	64/128/256
<i>Network Architecture</i>	
Encoder Network	3-layer MLP
Hidden Dimension	32
Activation Function	ReLU
Initialization	Standard Xavier

for general multimodal capability. In our PID analysis, MMBench tends to elicit synergy-dominant profiles across fusion-capable model families, indicating that visual and textual information are often jointly used in the final decision.

- **MMStar** (Chen et al., 2024b): MMStar is a vision-indispensable benchmark designed to reduce multimodal leakage and filter out samples that can be solved from language priors alone. It contains carefully validated multiple-choice questions where visual information is expected to be necessary for correct answering. Consistent with this design goal, our PID results show elevated $S_{v,l}$ on MMStar, making it a representative benchmark for studying cross-modal integration in vision–language models.
- **POPE** (Li et al., 2023): POPE is an object hallucination benchmark that probes whether LVLMs correctly judge the presence or absence of objects in an image through binary yes/no questions. Because the task directly tests object-level grounding, successful predictions often require matching textual object queries with visual evidence rather than relying solely on co-occurrence priors. In our analysis, POPE exhibits a grounding-sensitive, synergy-dominant profile, separating it from other hallucination-oriented benchmarks whose decisions may depend more strongly on language priors.

Regime II: Text-Unique / Prior-Dominant Knowledge Benchmarks. These benchmarks exhibit relatively high language-unique information (U_{txt}) in our experiments. Although they contain visual inputs, the evaluated models often rely strongly on textual context, answer priors, or domain knowledge when forming decisions.

- **MMMU (Massive Multi-discipline Multimodal Understanding)** (Yue et al., 2024): MMMU is a large-scale benchmark for college-level multidisciplinary multimodal understanding. It spans multiple disciplines and includes heterogeneous image types such as charts, diagrams, tables, chemical structures, and scientific illustrations. While visual interpretation is important for many questions, the benchmark also requires substantial domain knowledge and reasoning over specialized concepts. In our PID decomposition, MMMU tends to exhibit stronger U_{txt} , suggesting that model decisions are often influenced by language-side domain knowledge and answer priors.
- **PMC-VQA** (Zhang et al., 2023): PMC-VQA is a medical visual question-answering dataset derived from biomedical and clinical imagery, including modalities such as X-rays, MRIs, pathology images, and other medical figures. The questions often require specialized medical knowledge for interpretation and answer selection. Similar to MMMU, our analysis places PMC-VQA in a text-unique / prior-dominant regime, where language-side medical knowledge contributes substantially to the model’s predictive distribution.
- **Reefknot** (Zheng et al., 2025): Reefknot evaluates relational hallucination in LVLMs through multiple-choice questions involving spatial or semantic relationships among visual entities. Although it is related to hallucination evaluation, its

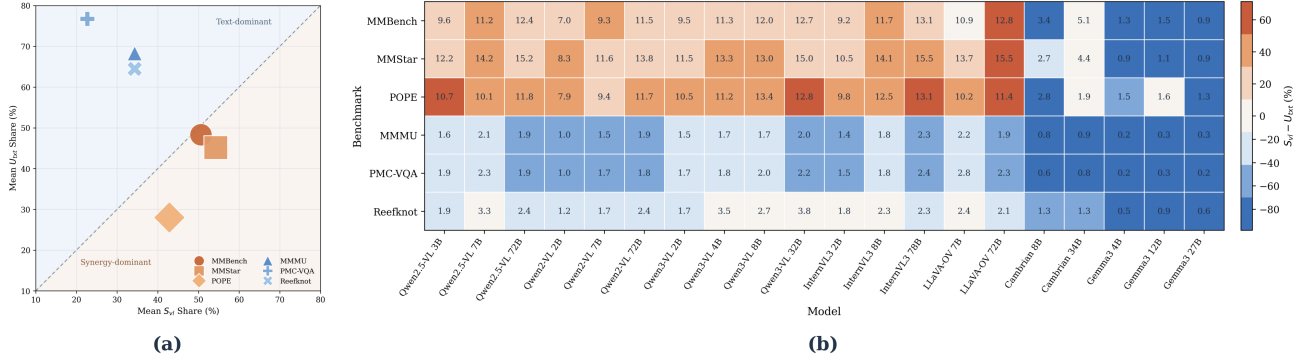


Figure 4. PID-derived modality-use analysis across 20 models and 6 vision-language benchmarks. (a) Benchmark landscape: each marker plots one benchmark’s mean S_{vl} share versus mean U_{txt} share, averaged across all 20 models. The dashed diagonal marks $S_{vl} = U_{txt}$; points above the diagonal exhibit relatively higher synergy, whereas points below exhibit relatively higher text-unique information. (b) Model-benchmark interaction profiles. Cell color denotes the profile gap (S_{vl} share - U_{txt} share, pp), with positive values indicating relatively stronger synergy and negative values indicating relatively stronger text-unique reliance. Cell values report $\Delta_{\text{vision}} = \text{Acc}(V, T) - \text{Acc}(T)$ (pp), where larger values indicate stronger behavioral dependence on vision.

interaction profile differs from POPE in our PID analysis. Whereas POPE focuses on object-level existence grounding, Reefknot often involves relational patterns that can be partially supported by textual priors or common visual-language associations. Our results show that Reefknot elicits stronger U_{txt} than POPE, illustrating that benchmarks with similar surface-level labels can induce distinct modality-use profiles.

E.2. Omni-Modal Benchmark

For the tri-modal analysis (V, A, T), we utilize a specific dataset capable of isolating audio-visual dependencies.

- **MUSIC-AVQA** (Li et al., 2022): A large-scale audio-visual dataset focused on music performance videos. The original dataset contains over 45K QA pairs encompassing spatial-temporal relations and acoustic comparisons.

Our Evaluation Split: To rigorously test the “Sensory Synergy Bottleneck,” we do not use the raw random split. Instead, we construct a balanced evaluation set consisting of three distinct subsets (1800 samples each) based on the dataset’s metadata:

1. **Audio-Focused:** Questions solvable primarily via auditory cues (e.g., identifying instruments by sound).
2. **Visual-Focused:** Questions relying on visual objects or actions (e.g., identifying visible instruments).
3. **AV-Fusion:** Questions explicitly requiring joint reasoning (e.g., “Is the violin playing the higher pitch?”). This subset serves as our critical testbed for identifying genuine sensory synergy (S_{av}).

F. Vision-Language Reference Analysis

This appendix reports the complete vision-language reference analyses summarized briefly in the main text. These experiments apply the analogous decision-level two-source PID formulation described in Appendix A.1 to 20 vision-language models across six benchmarks. Their purpose is to validate the implementation against previously reported LVLM PID behavior and to provide the vision-language diagnostics used by the reweighting study, rather than to constitute the primary empirical contribution of the paper.

F.1. Benchmark-Level Interaction Profiles

Figure 4(a) plots each benchmark’s mean S_{vl} share against its mean U_{txt} share, averaged across all 20 models. Consistent with the task-level separation reported by Xiu et al. (Xiu et al., 2026), the six-benchmark evaluation exhibits a similar division between relatively synergy-dominant and text-unique dominant profiles.

MMBench, MMStar, and POPE fall on the $S_{vl} > U_{txt}$ side of the diagonal, indicating relatively stronger cross-modal synergy, whereas MMMU, PMC-VQA, and Reefknot fall on the $U_{txt} > S_{vl}$ side, with U_{txt} shares exceeding 60%. We also recover

Table 6. Spearman ρ between PID quantities and two behavioral measures across 20 models. Each cell reports $\rho(\cdot, \Delta_{\text{vision}}) / \rho(\cdot, \text{Acc})$. I : total MI; CoI: co-information ($R_{\text{vl}} - S_{\text{vl}}$). $\ddagger p < .001$; $\dagger p < .01$; $* p < .05$; $^{\circ} p < .10$. Bold: $|\rho| \geq 0.68$.

Regime	Benchmark	S_{vl}	U_{txt}	U_{vis}	R_{vl}	$I(V, T; Y)$	CoI
Synergy-driven	MMBench	0.840[‡] / 0.752[‡]	-0.582 [†] / 0.198	0.068 / 0.058	0.098 / 0.088	-0.118 / 0.522 [*]	-0.822[‡] / -0.682[‡]
	MMStar	0.862[‡] / 0.762[‡]	-0.548 [*] / 0.148	0.042 / 0.032	0.062 / 0.052	-0.082 / 0.478 [*]	-0.850[‡] / -0.718[‡]
	POPE	0.798[‡] / 0.718[‡]	-0.502 [*] / 0.098	0.078 / 0.068	0.348 / 0.298	0.052 / 0.418 [°]	-0.722[‡] / -0.622[†]
Prior-dominant	MMMU	0.318 / 0.391	-0.198 / 0.422 [*]	0.038 / 0.058	0.048 / 0.042	-0.052 / 0.548 [*]	-0.298 / -0.242
	PMC-VQA	0.382 [°] / 0.398 [°]	-0.178 / 0.356 [*]	0.052 / 0.042	0.032 / 0.028	-0.078 / 0.593 [†]	-0.348 / -0.317
	Reefknot	0.418 [°] / 0.348	-0.278 / 0.378 [*]	0.068 / 0.072	0.058 / 0.048	0.022 / 0.552 [*]	-0.382 [°] / -0.278

the previously reported contrast between POPE and Reefknot (Xiu et al., 2026): despite both targeting hallucination-related behavior, POPE exhibits a more grounding-sensitive, synergy-heavy profile, whereas Reefknot shows stronger text-unique reliance. The additional MMStar and MMMU benchmarks follow the same broader separation.

Reference observation. Across the expanded benchmark set, the decision-level PID profiles broadly corroborate previously reported task-dependent modality-use regimes (Xiu et al., 2026), while extending the comparison to MMStar and MMMU.

F.2. Model–Benchmark Interaction Profiles

We next examine whether family-level modality-use tendencies persist across benchmarks and model scales. Consistent with prior LVLM PID analysis (Xiu et al., 2026), Figure 4(b) shows recurring differences across model families. Qwen2/2.5-VL, Qwen3-VL, InternVL3, and LLaVA-OneVision generally exhibit more *joint-modality-reliant* profiles, with higher S_{vl} and larger Δ_{vision} on synergy-dominant benchmarks, whereas Gemma3 and Cambrian tend toward stronger *language-prior-reliant* profiles with higher U_{txt} . These tendencies vary continuously across benchmarks rather than forming a strict dichotomy.

Scaling generally preserves these family-level tendencies. Notably, Qwen3-VL maintains a similar profile from 2B to 32B, while Gemma3 remains comparatively text-unique-heavy from 4B to 27B. Variation across benchmarks further indicates that these profiles are best interpreted as model–benchmark interactions rather than intrinsic properties of models alone.

Reference observation. Family-level modality-use tendencies broadly corroborate prior LVLM PID findings (Xiu et al., 2026) and persist in the updated evaluation, including Qwen3-VL and the additional MMStar and MMMU benchmarks.

F.3. Behavioral and Intervention Validity

We examine whether the reference PID profiles are associated with observable model behavior. For each model–benchmark pair, we define

$$\Delta_{\text{vision}} = \text{Acc}(V, T) - \text{Acc}(T), \quad (50)$$

where larger values indicate stronger behavioral dependence on vision.

Table 6 reports Spearman correlations between the PID quantities, accuracy, and Δ_{vision} across the 20 models for each benchmark.

PID components track regime-dependent behavior. Consistent with prior observations (Xiu et al., 2026), on MMBench, MMStar, and POPE, S_{vl} is strongly associated with both accuracy and vision-removal sensitivity:

$$\rho(S_{\text{vl}}, \Delta_{\text{vision}}) \geq 0.798, \quad \rho(S_{\text{vl}}, \text{Acc}) \geq 0.718 \quad (p < 0.001).$$

By contrast, U_{txt} is negatively associated with Δ_{vision} ($\rho \leq -0.502$, $p < 0.05$). On MMMU, PMC-VQA, and Reefknot, synergy is less dominant and text-unique information becomes relatively more prominent, while vision-removal effects remain small ($\Delta_{\text{vision}} < 3.5$ pp). MMStar and MMMU extend these regime-dependent patterns to additional benchmarks.

Total MI captures different behavior. Total mutual information $I(V, T; Y)$ correlates with accuracy ($\rho = 0.418$ – 0.593) but shows little association with vision-removal sensitivity ($|\rho| \leq 0.118$). Thus, total decision information and the decomposition of that information capture different aspects of model behavior.

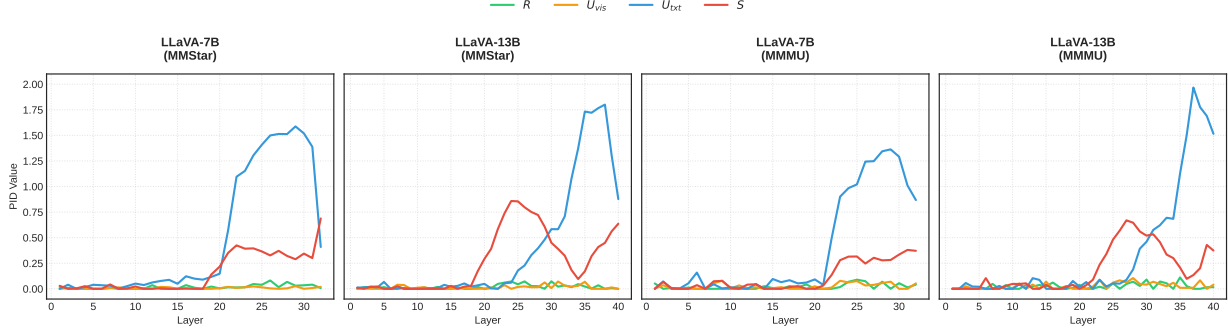


Figure 5. **Layer-wise PID in vision-language models.** PID atoms across transformer depth for LLaVA-1.5 (7B/13B) on MMStar and MMMU. Curves show Visual Uniqueness (U_{vis}), Language Uniqueness (U_{txt}), Redundancy (R_{vl}), and Synergy (S_{vl}). Depth is normalized to model layers.

Reference observation. Across six benchmarks, the PID-behavior relationships broadly corroborate prior LVLM findings (Xiu et al., 2026), with the validation extended to MMStar and MMMU.

F.4. Layer-Wise Vision-Language Reference

We analyze LLaVA-1.5 models on MMStar and MMMU as a layer-wise reference. Consistent with the three-phase dynamics reported in prior LVLM PID analysis (Xiu et al., 2026), Figure 5 exhibits a similar progression on these additional benchmarks.

Phase I: Weak decision-level information. During the early layers, all PID components remain small, indicating that little modality-specific or synergistic information is expressed in the intermediate decision distributions.

Phase II: Unimodal evidence accumulation. Across the middle layers, unique information—particularly U_{txt} —increases substantially, while S_{vl} remains comparatively limited or exhibits only a moderate intermediate rise.

Phase III: Late-stage fusion. Toward the output layers, S_{vl} increases sharply while U_{txt} declines, reproducing the late-stage shift from language-unique information toward cross-modal synergy reported previously (Xiu et al., 2026).

The strength of this transition varies with the benchmark profile. On MMStar, the final synergy increase is pronounced and can exceed U_{txt} , whereas on MMMU, text-unique information remains comparatively stronger and the late synergy response is weaker. Thus, while the broad three-phase organization persists across these benchmarks, its final balance reflects task-dependent modality use.

Reference observation. The previously reported three-phase layer-wise PID pattern also appears on MMStar and MMMU, with stronger late-stage synergy on the more synergy-dominant benchmark.

F.5. Full Per-Model PID and Intervention Results

Tables 7 and 8 report the complete PID decomposition and vision-removal intervention results for all 20 models across six benchmarks. Each cell contains the four PID terms (S_{vl} , U_{txt} , U_{vis} , R_{vl}), full multimodal accuracy (Acc), text-only accuracy (A_{t}), and the vision-removal drop $\Delta = \text{Acc} - A_{\text{t}}$ (pp). Models are grouped by the descriptive family-level modality-use profiles discussed above: blue rows denote relatively joint-modality-reliant families and red rows denote relatively language-prior-reliant families. All PID values are in bits; “ ∇ ” denotes < 0.01 bits.

The full tables provide the underlying data for Figure 4 and the Spearman correlations in Table 6.

Cross-table patterns. Several structural regularities are visible across the full model set. (i) U_{vis} is **uniformly small** (< 0.03 bits across all model-benchmark pairs). (ii) R_{vl} is **format-dependent**: redundancy remains below 0.02 bits on the four-option MCQ benchmarks but rises on POPE, whose binary answer space permits greater overlap between vision- and text-derived evidence. (iii) **Family-level profiles are largely preserved across scale**: for example, Qwen3-VL

maintains $S_{vl} > U_{txt}$ on the synergy-dominant benchmarks from 2B to 32B, whereas Gemma3 remains strongly U_{txt} -heavy from 4B to 27B. (iv) **Profile-intervention coherence**: relatively joint-modality-reliant models exhibit larger Δ_{vision} on synergy-dominant benchmarks, whereas language-prior-reliant models exhibit substantially smaller vision-removal effects.

Tables 7 and 8 report the complete PID decomposition and vision-removal intervention results for all 20 models across 6 benchmarks. Each cell contains the four PID terms (S_{vl} , U_{txt} , U_{vis} , R_{vl}), full multimodal accuracy (Acc), text-only accuracy (A_t), and the vision-removal drop $\Delta = \text{Acc} - A_t$ (pp). Models are grouped by family-level modality-use profile: blue rows denote joint-modality-reliant families and red rows denote language-prior-reliant families, as identified via the PID analysis in §F.1. All PID values are in bits; “ ∇ ” denotes < 0.01 bits.

These tables extend the representative subset shown in Figure 4 (b) to the full model set, and provide the underlying data for all Spearman correlations reported in Table 6.

Cross-table patterns. Several structural regularities are visible across the full model set. (i) U_{vis} is **uniformly small** (< 0.03 bits across all model–benchmark pairs), confirming that vision-unique information plays a negligible role in MCQ decision-making. (ii) R_{vl} is **format-dependent**: redundancy remains below 0.02 bits on all four-option MCQ benchmarks but rises to 0.18–0.42 bits on POPE (binary Yes/No format), where the reduced option space allows greater overlap between vision-derived and text-derived evidence. (iii) **Family-level profiles are scale-invariant**: within each family, increasing model scale amplifies the dominant PID term without switching the qualitative profile. For example, Qwen3-VL maintains $S_{vl} > U_{txt}$ on synergy-driven benchmarks from 2B to 32B, while Gemma3 maintains $U_{txt} \gg S_{vl}$ from 4B to 27B. (iv) **Profile-intervention coherence**: joint-modality-reliant models show $\Delta_{\text{vision}} = 7\text{--}15$ pp on synergy-driven benchmarks but < 3.5 pp on prior-dominant benchmarks; language-prior-reliant models show $\Delta_{\text{vision}} < 5.1$ pp even on synergy-driven benchmarks. This pattern supports the interpretation that PID profiles capture functionally meaningful modality dependence (§F.3).

G. Ablation Studies and Sensitivity Analysis

To validate the robustness of our Sensory PID methodology, we examine the sensitivity of the estimated information components to two critical implementation choices: (i) the strategy for feature summarization (Mean Pooling vs. Last Hidden State vs. Max Pooling), and (ii) the confidence gating threshold $\tau \in \{0.3, 0.4, 0.5\}$.

Table 9 presents the results on MMBench for three representative models. We report the values for Synergy and Text Uniqueness (S_{vl}/U_{txt}). The results demonstrate that our estimates are highly stable across different feature aggregation methods and confidence thresholds, with variations remaining negligible (typically < 0.01 bits). This indicates that the observed interaction profiles are robust to these estimation choices rather than artifacts of the selected hyperparameters.

H. PID-Guided Reweighting: Full Details

This section provides complete implementation details, validation experiments, ablation studies, and supplementary results for the PID-guided reweighting in §5. The main text presents the core results (Tables 2); here we report all supporting evidence for reproducibility and interpretation.

H.1. Algorithm and Experimental Conditions

Algorithm 1 summarizes the PID-guided sample reweighting sample selection procedure described in §2.3. Table 10 provides full descriptions of all six experimental conditions.

Algorithm 1 PID-Guided Sample Selection

Require: Per-sample scores $\{\text{SC}_i, \text{GapScore}_i\}_{i=1}^N$; selection counts K_s, K_g

- 1: $\mathcal{S}_{\text{short}} \leftarrow \text{TopK}_{K_s}(\text{SC}_i)$ {Priority 1: language-shortcut}
 - 2: $\mathcal{R} \leftarrow \{1, \dots, N\} \setminus \mathcal{S}_{\text{short}}$
 - 3: $\mathcal{S}_{\text{gap}} \leftarrow \text{TopK}_{K_g}(\text{GapScore}_i \mid i \in \mathcal{R})$ {Priority 2: under-synergized}
 - 4: $\mathcal{S}_{\text{normal}} \leftarrow \mathcal{R} \setminus \mathcal{S}_{\text{gap}}$
 - 5: Assign weights: $w_i = 3.0$ if $i \in \mathcal{S}_{\text{gap}}$; $w_i = 0.5$ if $i \in \mathcal{S}_{\text{short}}$; $w_i = 1.0$ otherwise
-

Table 7. Full PID decomposition and intervention results on **synergy-driven benchmarks** (20 models $\times$ 3 benchmarks). For each model–benchmark pair: S_{vl} , U_{txt} , U_{vis} , R_{vl} are PID terms (bits); Acc = multimodal accuracy (%); A_t = text-only accuracy (%); Δ = Acc – A_t (pp). Blue = joint-modality-reliant; red = language-prior-reliant. “ ∇ ” = <0.01 bits.

Model	Size	MMBench							MMStar							POPE						
		S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ	S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ	S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ
Qwen2.5-VL	3B	.88	.52	.01	.01	78.1	68.5	9.6	.80	.45	.01	∇	52.4	40.2	12.2	.54	.15	.02	.22	84.9	74.2	10.7
Qwen2.5-VL	7B	1.10	.63	∇	.01	85.4	74.2	11.2	1.12	.58	.02	∇	60.7	46.5	14.2	.57	.23	.02	.35	86.4	76.3	10.1
Qwen2.5-VL	72B	1.09	.83	∇	.02	91.9	79.5	12.4	1.12	.80	.01	∇	68.2	53.0	15.2	.58	.20	.01	.18	89.6	77.8	11.8
Qwen2-VL	2B	.85	.52	.01	.01	72.5	65.5	7.0	.72	.38	.01	∇	44.8	36.5	8.3	.52	.18	.02	.27	82.4	74.5	7.9
Qwen2-VL	7B	1.12	.52	.01	.01	81.3	72.0	9.3	.62	.50	.01	∇	56.9	45.3	11.6	.45	.28	.02	.30	85.2	75.8	9.4
Qwen2-VL	72B	.86	.66	∇	.02	88.5	77.0	11.5	.98	.70	.01	∇	62.8	49.0	13.8	.55	.22	.01	.36	88.2	76.5	11.7
Qwen3-VL	2B	.72	.48	.01	.01	76.5	67.0	9.5	.74	.42	.01	∇	48.5	37.0	11.5	.46	.24	.02	.30	83.5	73.0	10.5
Qwen3-VL	4B	.95	.55	∇	.01	82.8	71.5	11.3	.98	.48	.01	∇	55.8	42.5	13.3	.52	.22	.02	.33	86.2	75.0	11.2
Qwen3-VL	8B	1.08	.58	∇	.01	86.0	74.0	12.0	1.05	.52	.01	∇	59.0	46.0	13.0	.55	.22	.02	.35	89.2	75.8	13.4
Qwen3-VL	32B	1.22	.75	∇	.02	90.5	77.8	12.7	1.25	.70	.01	∇	65.5	50.5	15.0	.62	.20	.01	.38	89.8	77.0	12.8
InternVL3	2B	.55	.38	.01	.01	81.7	72.5	9.2	.52	.35	.01	∇	50.5	40.0	10.5	.48	.25	.02	.28	85.3	75.5	9.8
InternVL3	8B	.88	.42	∇	.01	88.0	76.3	11.7	.80	.35	∇	∇	62.3	48.2	14.1	.55	.22	.01	.35	88.5	76.0	12.5
InternVL3	78B	1.15	.70	∇	.02	91.1	78.0	13.1	1.38	.48	.01	∇	67.5	52.0	15.5	.62	.18	.01	.40	89.6	76.5	13.1
LLaVA-OV	7B	.97	.85	.01	.01	84.7	73.8	10.9	.95	.60	.01	∇	58.7	45.0	13.7	.52	.25	.02	.32	85.7	75.5	10.2
LLaVA-OV	72B	1.47	.29	∇	.01	91.3	78.5	12.8	1.50	.25	∇	∇	66.8	51.3	15.5	.65	.18	.01	.42	89.2	77.8	11.4
Cambrian	8B	.05	1.47	.01	.01	71.6	68.2	3.4	.38	.55	.01	∇	46.5	43.8	2.7	.03	.65	.02	.25	79.3	76.5	2.8
Cambrian	34B	.62	.82	.01	.01	79.6	74.5	5.1	.55	.68	.01	∇	52.4	48.0	4.4	.42	.38	.02	.28	86.1	84.2	1.9
Gemma3	4B	.17	1.70	∇	.01	78.8	77.5	1.3	.15	1.42	∇	∇	49.4	48.5	0.9	.26	.74	.01	.30	82.0	80.5	1.5
Gemma3	12B	.19	1.75	∇	.02	82.5	81.0	1.5	.18	1.68	∇	∇	54.8	53.7	1.1	.53	.49	.01	.35	82.1	80.5	1.6
Gemma3	27B	.15	1.78	∇	.01	84.1	83.2	0.9	.14	1.72	∇	∇	56.7	55.8	0.9	.11	.78	∇	.22	83.8	82.5	1.3

Table 8. Full PID decomposition and intervention results on **prior-dominant benchmarks** (20 models $\times$ 3 benchmarks). Format and conventions follow Table 7.

Model	Size	MMMU							PMC-VQA							Reefknot						
		S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ	S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ	S_{vl}	U_{txt}	U_{vis}	R_{vl}	Acc	A_t	Δ
Qwen2.5-VL	3B	.28	.60	.01	∇	41.4	39.8	1.6	.25	.55	.01	∇	41.4	39.5	1.9	.45	.65	.01	.01	52.4	50.5	1.9
Qwen2.5-VL	7B	.35	.72	∇	∇	54.1	52.0	2.1	.32	.65	∇	∇	53.1	50.8	2.3	.62	.59	.01	.01	65.4	62.1	3.3
Qwen2.5-VL	72B	.28	.88	∇	∇	62.4	60.5	1.9	.26	.89	∇	∇	58.4	56.5	1.9	.52	.82	∇	.01	74.2	71.8	2.4
Qwen2-VL	2B	.15	.48	.01	∇	34.0	33.0	1.0	.12	.45	.01	∇	34.5	33.5	1.0	.30	.55	.01	.01	42.0	40.8	1.2
Qwen2-VL	7B	.22	.65	.01	∇	44.0	42.5	1.5	.20	.60	.01	∇	44.0	42.3	1.7	.38	.62	.01	.01	55.2	53.5	1.7
Qwen2-VL	72B	.28	.78	∇	∇	58.4	56.5	1.9	.25	.72	∇	∇	52.8	51.0	1.8	.48	.72	∇	.01	68.4	66.0	2.4
Qwen3-VL	2B	.25	.55	.01	∇	38.5	37.0	1.5	.22	.50	.01	∇	38.2	36.5	1.7	.40	.60	.01	.01	48.5	46.8	1.7
Qwen3-VL	4B	.30	.62	∇	∇	45.5	43.8	1.7	.28	.58	∇	∇	45.0	43.2	1.8	.50	.62	.01	.01	55.5	52.0	3.5
Qwen3-VL	8B	.32	.68	∇	∇	50.5	48.8	1.7	.30	.62	∇	∇	49.5	47.5	2.0	.55	.65	.01	.01	63.9	61.2	2.7
Qwen3-VL	32B	.32	.82	∇	∇	60.5	58.5	2.0	.30	.78	∇	∇	57.2	55.0	2.2	.58	.75	∇	.01	72.0	68.2	3.8
InternVL3	2B	.22	.55	.01	∇	43.4	42.0	1.4	.20	.52	.01	∇	42.0	40.5	1.5	.42	.50	.01	.01	52.0	50.2	1.8
InternVL3	8B	.26	.58	∇	∇	57.4	55.6	1.8	.25	.55	∇	∇	52.0	50.2	1.8	.50	.55	∇	.01	64.1	61.8	2.3
InternVL3	78B	.28	.82	∇	∇	68.8	66.5	2.3	.26	.78	∇	∇	57.4	55.0	2.4	.48	.72	∇	.01	68.8	66.5	2.3
LLaVA-OV	7B	.32	.45	.01	∇	53.4	51.2	2.2	.30	.42	.01	∇	54.8	52.0	2.8	.55	.58	.01	.01	56.9	54.5	2.4
LLaVA-OV	72B	.30	.87	∇	∇	59.4	57.5	1.9	.28	.85	∇	∇	56.8	54.5	2.3	.48	.82	∇	.01	70.1	68.0	2.1
Cambrian	8B	.08	.58	.01	∇	43.0	42.2	0.8	.06	.55	.01	∇	39.6	39.0	0.6	.12	.65	.01	∇	46.5	45.2	1.3
Cambrian	34B	.10	.72	.01	∇	49.4	48.5	0.9	.08	.70	.01	∇	46.4	45.6	0.8	.18	.82	.01	∇	52.4	51.1	1.3
Gemma3	4B	.02	1.75	∇	∇	47.2	47.0	0.2	.02	1.86	∇	∇	43.4	43.2	0.2	.12	1.45	∇	∇	40.5	40.0	0.5
Gemma3	12B	.03	1.85	∇	∇	54.6	54.3	0.3	.03	1.95	∇	∇	51.4	51.1	0.3	.16	1.55	∇	∇	46.7	45.8	0.9
Gemma3	27B	.02	1.90	∇	∇	57.4	57.1	0.3	.02	2.00	∇	∇	53.4	53.2	0.2	.14	1.60	∇	∇	49.8	49.2	0.6

Table 9. Ablation study of PID components (S_{vl}/U_{txt}) on MMBench. We compare sensitivity to different feature summarization strategies (left) and confidence thresholds τ (right). The estimates remain consistent across settings.

Model	Representation Feature			Confidence Threshold (τ)		
	Mean Pooling	Last Hidden	Max Pooling	0.3	0.4	0.5
Qwen2.5-VL-7B	1.1012/0.6338	1.1024/0.6359	1.1018/0.6333	1.1012/0.6338	1.1017/0.6336	1.1018/0.6381
Gemma3-12B	0.1912/1.7513	0.1908/1.7516	0.1926/1.7521	0.1912/1.7513	0.1912/1.7548	0.1912/1.7598
LLaVA-ov-7B	0.9736/0.6321	0.9737/0.6326	0.9735/0.6311	0.9736/0.6321	0.9736/0.6324	0.9742/0.6344

Table 10. Six experimental conditions. Conditions C–F use matched counts: $K_g = 389$ samples at $3.0\times$ and $K_s = 746$ at $0.5\times$.

ID	Condition	Description
A	Base	Frozen pretrained model. Zero-shot reference.
B	LoRA-Uniform	Standard LoRA, all samples $w = 1.0$. Isolates fine-tuning gain.
C	LoRA-PID	Top- K_g by GapScore (Eq. 11) $\uparrow$; top- K_s by SC $\downarrow$.
D	LoRA-Random	Same weight distribution as C, randomly assigned. Controls for non-uniformity.
E	LoRA-Acc	Top- K_g by difficulty (incorrect samples ranked first; ties broken by $H(p_{vt}^{(i)})$); top- K_s easiest (correct with lowest entropy).
F	LoRA-Ablation	Top- K_g by $\text{AblScore}_i = H(p_t^{(i)}) - H(p_{vt}^{(i)})$ (joint-over-text entropy reduction); top- K_s by lowest AblScore. Tests whether simple entropy reduction from adding vision suffices without distinguishing unique vs. synergistic contributions.

H.2. LoRA Configuration and Data Controls

LoRA configuration. All conditions (B–F) use identical LoRA settings: rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, learning rate 2×10^{-5} with cosine schedule and 5% warmup, effective batch size 16. LoRA adapters target in the last 20% of each model’s transformer layers. This targeting is motivated by the layer-wise analysis (§??), which shows cross-modal synergy emerging in the final layers. Per-model layer ranges: Qwen2.5-VL-7B (layers 23–28 of 28), LLaVA-OV-7B (layers 26–32 of 32), Gemma3-12B (layers 25–30 of 30). Three random seeds {42, 123, 456} are used for all conditions; Condition D uses 3 random weight assignments per seed (9 runs total).

No-leakage guarantee. No evaluation samples from MMBench-test, MMStar, POPE, MMMU, or PMC-VQA are used for PID profiling, hyperparameter selection, or LoRA training.

Evaluation protocol. Post-tuning PID diagnostics (Post- S_{vl} , Post- U_{txt}) are estimated on a held-out MMStar eval subset via the BATCH estimator using the same hyperparameters as the main analysis. Base PID values are consistent with the main paper. All accuracy results are reported as mean $\pm$ std over 3 seeds. Confidence intervals are computed via seed-stratified paired bootstrap (10,000 resamples).

H.3. Effective Training Distribution

Reweighting scheme. Table 11 reports the effective training distribution.

Table 11. Reweighting scheme and effective training distribution.

Category	Count	Raw %	Weight	Eff. Wt.	Eff. %
Synergy-Gap	389	13%	$3.0\times$	1,167	34.0%
Normal	1,865	62%	$1.0\times$	1,865	54.9%
Shortcut	746	25%	$0.5\times$	373	11.1%
Total	3,000	100%	—	3,405	100%

H.4. Supplementary Results

Pairwise comparisons with confidence intervals. Table 12 reports all pairwise deltas on MMStar with 95% paired bootstrap CIs (seed-stratified; 10,000 resamples).

Table 12. Pairwise comparisons on MMStar with 95% paired bootstrap CIs.

Comparison	ΔMMStar	95% CI	$\Delta\text{Post-}S_{\text{vl}}$	Interpretation
C vs B	+2.3	[1.4, 3.2]	+0.16	PID > standard fine-tuning
C vs D	+2.8	[1.6, 4.0]	+0.20	PID signal > random assignment
C vs E	+1.8	[0.7, 2.9]	+0.14	PID > difficulty-based
C vs F	+1.3	[0.2, 2.4]	+0.12	PID > ablation-based
C vs A	+3.6	[2.7, 4.5]	+0.21	Overall PID-guided gain

Per-category accuracy. Table 13 reports per-category results on the MMBench test set, showing regime-selective gains.

Table 13. Per-category accuracy on MMBench test (Qwen2.5-VL-7B). $\Delta_{\text{PID}} = \text{LoRA-PID} - \text{LoRA-Uniform}$.

Subtype	Regime	LoRA-Base	LoRA-Unif.	LoRA-PID	LoRA-Acc	LoRA-Abl.	Δ_{PID}
Spatial Reasoning	Reasoning	82.1	84.0	87.5	84.8	85.5	+3.5
Attr. Comparison	Reasoning	84.5	86.2	89.0	86.8	87.3	+2.8
Action Recognition	Reasoning	85.8	87.0	89.5	87.5	88.0	+2.5
Object Localization	Percep.	91.2	92.0	93.1	92.3	92.6	+1.1
Scene Classification	Percep.	90.5	91.0	91.5	91.2	91.3	+0.5
OCR / Text Recog.	Knowl.	89.0	89.5	89.2	89.8	89.4	−0.3
Commonsense	Knowl.	88.3	88.8	88.5	89.2	88.7	−0.3

POPE subset breakdown. Table 14 shows PID-guided gains are largest on POPE-Adversarial (+1.9 over Uniform), consistent with shortcut reduction mitigating co-occurrence-based hallucination priors.

Table 14. POPE subset results (Qwen2.5-VL-7B).

Cond.	Random	Popular	Adversarial	Avg
Base	88.2	86.5	84.5	86.4
LoRA-Uniform	88.8	87.2	85.6	87.2
LoRA-PID	89.6	88.3	87.5	88.5
LoRA-Acc-RW	89.0	87.5	86.0	87.5
LoRA-Abl-RW	89.2	87.7	86.5	87.8

PID profile shift. Table 15 reports the full post-tuning PID profile on the held-out MMStar eval subset.

Table 15. Post-tuning PID profile (MMStar eval subset, Qwen2.5-VL-7B).

Condition	S_{vl}	U_{txt}	U_{vis}	R_{vl}	S_{vl} Share	ΔAcc
Base	1.15	0.60	0.015	0.002	65.1%	7.7
LoRA-Uniform	1.20	0.56	0.016	0.002	67.5%	8.2
LoRA-PID	1.36	0.46	0.018	0.003	73.9%	10.8
LoRA-Acc-RW	1.22	0.54	0.016	0.002	68.3%	8.6
LoRA-Abl-RW	1.24	0.52	0.017	0.002	69.3%	9.0

H.5. Cross-Model Full Results

Table 16 provides full results for LLaVA-OneVision-7B and Gemma3-12B.

Table 16. Full cross-model results (mean $\pm$ std over 3 seeds).

Model	Cond.	MMBench	MMStar	MMMU	P- S_{vl}	P- U_{txt}
LLaVA-OV-7B	LoRA-Base	84.7	58.7	53.4	0.95	0.61
	LoRA-Uniform	85.5 $\pm$.3	60.0 $\pm$.4	53.1 $\pm$.3	1.00 $\pm$.02	0.58 $\pm$.02
	LoRA-PID	86.3$\pm$.3	61.5$\pm$.4	52.8 $\pm$.4	1.10$\pm$.03	0.50$\pm$.02
Gemma3-12B	LoRA-Base	82.5	54.8	54.6	0.18	1.68
	LoRA-Uniform	83.1 $\pm$.4	56.2 $\pm$.5	59.3 $\pm$.3	0.20 $\pm$.02	1.65 $\pm$.03
	LoRA-PID	83.3 $\pm$.4	54.3 $\pm$.5	54.1 $\pm$.4	0.21 $\pm$.02	1.63 $\pm$.03

The Gemma3-12B negative case (+0.3, within seed variation) suggests PID-guided reweighting amplifies existing fusion capacity rather than creating it, reinforcing the diagnostic utility of the modality-use profiles (§F.1).

H.6. Ablation Studies

All ablations are conducted on Qwen2.5-VL-7B.

Upweight factor sensitivity. Table 17 varies the upweight factor applied to synergy-gap samples ($K_g = 389$, $K_s = 750$, downweight = $0.5\times$). Performance peaks at $3\times$ and plateaus at $5\times$, while MMMU shows monotonic decrease.

Table 17. Upweight factor sensitivity.

Upweight	MMStar	Post- S_{vl}	MMMU
$1\times$ (= Uniform)	62.0	1.20	54.0
$2\times$	63.2	1.28	53.8
$3\times$	64.3	1.36	53.5
$5\times$	64.0	1.40	52.8

Selection count sensitivity. Table 18 varies the shortcut selection count K_s . Setting K_s too aggressively (40%) hurts knowledge benchmarks more than it helps reasoning; being too conservative (10%) limits impact.

Table 18. Selection count sensitivity (K_s as % of N , upweight = $3\times$).

K_s (% of N)	Downweighted	MMStar	MMMU
40%	$\sim 1,200$	63.8	52.9
25%	~ 750	64.3	53.5
10%	~ 300	63.0	53.9

H.7. Per-Seed Results

Table 19 confirms that LoRA-PID achieves the highest MMStar accuracy across all three seeds individually, with a worst-case advantage of +1.3 over LoRA-Ablation (Seed 123).

Table 19. Per-seed results on MMStar and MMMU (Qwen2.5-VL-7B). Condition D: mean over 3 random assignments per seed.

Condition	MMStar			MMMU		
	Seed 42	Seed 123	Seed 456	Seed 42	Seed 123	Seed 456
B LoRA-Uniform	62.4	61.7	61.9	54.2	53.8	54.0
C LoRA-PID	64.6	64.0	64.3	53.2	53.9	53.4
D LoRA-Random	61.8	61.2	61.5	54.3	53.7	54.3
E LoRA-Acc	62.9	62.2	62.4	54.5	54.0	54.4
F LoRA-Ablation	63.4	62.7	62.9	54.0	53.5	53.9